



Utilising a New Improved Multi-Layer Perceptron-based Artificial Gorilla Troops Optimiser Algorithm for Biomedical Image Analysis to Detect Lung Cancer

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Received: 18 April 2025 / Accepted: 24 May 2026

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Abstract

Timely clinical intervention relies on the early as well as precise identification of lung cancer from histopathological images. This research introduces an Enhanced Multilayer Perceptron optimized via Artificial Gorilla Troops Optimization (EMLP-AGTO), designed as a streamlined as well as computationally effective substitute for traditional deep CNN methods. The system combines Wiener filtering to reduce noise as well as Local Tetra-Pattern (LTrP) descriptors to extract distinctive texture signatures from H&E-stained slides. AGTO dynamically adjusts MLP parameters to ensure stable convergence as well as improve classification dependability. Tests conducted on the LC25000 dataset show robust as well as reliable effectiveness, attaining 99.89% accuracy, 99.86% precision, 99.91% sensitivity, and a 99.88% F1-score through five-fold cross-validation. Benchmarking additionally indicates that EMLP-AGTO trains 6–7 times quicker as well as achieves an inference speed of 2.1 ms per image, surpassing VGG16 and ResNet50 in computational efficiency. These findings emphasize the practical applicability of the suggested method for quick initial assessment in digital pathology processes. In summary, EMLP-AGTO functions as an interpretable as well as resource-effective tool for decision support that can enhance diagnostic efficiency in clinical environments.

Keywords Biomedical image analysis · Lung cancer detection · Enhanced multi-layer perceptron · Artificial Gorilla Troops Optimizer Algorithm · Wiener filter · Local tetra pattern

1 Introduction

Cancer continues to be a major health issue worldwide, with lung cancer being the primary reason for deaths associated with cancer. Timely identification as well as accurate diagnosis are crucial for lowering death rates, but the difficulty of detecting lung cancer in its initial phases presents obstacles. Traditional approaches depend greatly on radiographic imaging as well as manual evaluation of histopathology images, which are labor-intensive, prone to human mistakes, and frequently limit the required accuracy for timely intervention. Furthermore, differences in how medical professionals interpret images add complexity to diagnosis as well as treatment planning, hindering the establishment of consistent and dependable detection standards. The unchecked advancement of aberrant cells from the body's organs or tissues causes cancer (Su et al. 2022). Cancer cells may then spread to other bodily organs or tissues. Smoking negatively impacts the development of lung tumours. Muscle soreness,

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exhaustion, coughing, and other symptoms are typical, and they are followed by a variety of syndromes (Nazir et al. 2023). To identify cancer, radiographic imaging techniques are often employed (Yang et al. 2023). Out of these, histopathological images containing phenotypic data are crucial for evaluating and diagnosing cancer diseases (Patel et al. 2022). It outlines a difficult and delicate task for experts to individually assess these medical images. This procedure is labor-intensive as well as requires considerable focus from experts. When initially identified, it becomes harder to recognize patients because the symptoms are subtle as well as complicated to define (Ahmed et al. 2023).

Similarly, the process of diagnosing tumors is time-consuming and initially contingent on the differing opinions of medical professionals (Mansour et al. 2022). A different area of medicine can resolve these issues. AI applications in medicine encompass disease prediction and biomedical image analysis, as well as early detection of health crises (Assad and Kiczales 2020). DL techniques can analyze high-dimensional images and videos, as well as anatomical descriptions, with remarkable accuracy (Alanazi et al. 2022). In an identical manner, Deep Learning (DL) methods are utilized to detect tumors at an early stage as well as differentiate among various phases of the illness by capturing both features and latent characteristics (Obayya et al. 2022).

The management of intricate DL system configurations and behaviors can be achieved through the use of multi-tiered hyperparameters, referred to as hierarchical hyperparameters (VanHorn and Çobanoğlu 2022). In this hierarchical structure of hyperparameters, higher-level hyperparameters affect the ranges or values linked to lower-level hyperparameters (Pradhan et al. 2023). By offering a structured approach to managing as well as adjusting each component's parameters, it simplifies the overall process, making it easier to understand and modify (Barzekar and Yu 2022). The hierarchical relationship between a feature extractor and a classifier must be considered when optimizing hyperparameters for these elements (Habib and Qureshi 2020). Finding the perfect equilibrium between generalization and method complexity can be enhanced by adjusting this hyperparameter through Bayesian optimization or random search (Devarajan et al. 2024; Manoharan et al. 2021; Rekha et al. 2022; Nuthal Srinivasan et al. 2024).

Recent progress in Artificial Intelligence (AI) as well as DL has enabled the creation of automated frameworks that can process biomedical images effectively and with great precision. Current AI-driven techniques, such as CNNs as well as Recurrent Neural Networks (RNNs), have demonstrated potential in a range of diagnostic uses. Nonetheless, these methods typically need considerable computational resources as well as extended training durations, and their effectiveness might decline with inferior-quality images

or intricate biological textures. Conventional optimization methods employed in these methods, including Bayesian optimization or random search, face constraints in effectively adjusting hyperparameters to improve effectiveness.

Additionally, approaches like Hybrid Metaheuristics with DL-oriented Fusion Model (HMDL-MFMBIA), Fuzzy C-Means (FCM), as well as Multi-Scale Convolutional Network (MSConvNeXt), have been suggested to tackle the challenges in biomedical image segmentation and classification. Although these techniques have demonstrated enhancements in certain instances, they continue to encounter challenges regarding the equilibrium between precision and computational effectiveness, particularly with large biomedical datasets. Low image quality as well as inadequate feature extraction methods continue to be major obstacles, frequently leading to less than ideal effectiveness.

In response to these issues, this study presents a new EMLP-AGTO algorithm. The EMLP-AGTO method utilizes the group dynamics of gorilla bands to effectively tune parameters of MLP, surpassing the constraints of conventional optimization methods. The suggested method employs the LC25000 dataset, a well-known standard for lung cancer image assessment, as well as improves image pre-processing with a Wiener filter to minimize noise and enhance image quality. The LTrP feature extractor is subsequently utilized to seize spatial as well as texture information, which is essential for precise classification.

Through the incorporation of these enhanced techniques, the EMLP-AGTO algorithm exhibits enhanced effectiveness regarding accuracy and precision, as well as sensitivity when measured against current methods like HMDL-MFMBIA, FCM, and MSConvNeXt. It tackles essential challenges related to computational complexity and fluctuations in image quality, as well as effectiveness in optimization, providing a scalable and reliable solution for lung cancer detection. This innovative methodology could enhance early detection and minimize manual involvement as well as establish a novel benchmark for AI-based biomedical image analysis in clinical settings.

The main improvements of this study are as follows:

- **Development of a Novel EMLP-AGTO Algorithm:** A novel algorithm rooted in the AGTO has been created to fine-tune MLP parameters, boosting computational effectiveness as well as improving classification accuracy in lung cancer detection.
- **Enhanced Image Pre-Processing Technique:** A Wiener filter is applied in image pre-processing to minimize noise and boost contrast, as well as improve overall image quality, resulting in improved feature extraction.
- **Implementation of Advanced Feature Extraction Using LTrP:** The LTrP method is utilized to extract

complex spatial as well as texture attributes from biomedical images, enhancing the method's capacity to distinguish between lung benign tissue, lung adenocarcinoma, and lung squamous cell carcinoma.

- **Comprehensive Performance Evaluation:** The EMLP-AGTO algorithm is tested against current methodologies (HMDL-MFMBIA, FCM, BDA-BMIC, as well as MSConvNeXt), showcasing better effectiveness regarding accuracy, precision, and sensitivity.
- **Scalability and Clinical Applicability:** The study indicates the possibility of utilizing the EMLP-AGTO method for detecting various cancer kinds with distinct datasets, emphasizing its scalability as well as applicability for practical clinical uses.

1.1 Motivation

Modern medicine has seen tremendous advancements in medical imaging, which can presently record a range of biological images from patients in order to assess as well as evaluate the condition along with its severity. Computer vision and AI have the potential to detect hidden data in biological images and to do better than human diagnosis. Reliability as well as quick prediction are critical factors in healthcare applications that provide early illness identification. The current frameworks are more sophisticated, require longer calculation times, and have less classification accuracy. Poor image quality might affect the processing technique and produce less than ideal outcomes. Moreover, comprehensive pre-processing methods are required to provide precise results. One of the major important visual factors is image contrast. Inadequate contrast can pose several difficulties for computer vision methods. For many cases, existing contrast adjustment approaches might not be sufficient. Sometimes, these methodologies produce images that are devoid of important details. The use of biomedical image processing in computer-oriented diagnosis is essential. Computer-aided diagnostic technologies play a major role in the current medical care industry. The development of computer-assisted diagnosis in healthcare depends heavily on the discipline of biomedical image processing. This kind of image enhancement allows doctors to view more details and diagnose patients with more knowledge. The capacity of biomedical image processing algorithms to identify and localize lesions or abnormalities facilitates early cancer diagnosis.

1.2 Problem Statement

Lung cancer, the leading cause of cancer-related deaths worldwide, requires early and accurate detection to improve

survival rates significantly. However, current diagnostic practices face several challenges:

- Conventional techniques rely significantly on radiographic imaging as well as manual analysis of histopathology images, resulting in diagnostic variability stemming from human subjectivity and differing interpretations. These manual evaluations frequently take a lot of time, leading to postponements in treatment.
- Biomedical images often experience noise and low contrast, as well as inadequate resolution, which affects the efficiency of image processing methods. Current methods, although somewhat efficient, frequently do not fully tackle these image quality problems, leading to inconsistent as well as erroneous diagnoses.
- Computationally intensive resources as well as prolonged training periods are necessary for deep learning methods such as CNNs and hybrid techniques like HMDL-MFMBIA. Moreover, conventional optimization methods such as Bayesian optimization as well as random search lack the efficiency necessary for hyperparameter tuning aimed at enhancing method effectiveness in these intricate configurations.
- Current methods, such as FCM, BDA-BMIC, and MSConvNeXt, exhibit shortcomings in reaching the high precision and sensitivity required for clinical dependability. Even with efforts to incorporate sophisticated algorithms, these methods find it challenging to achieve a balance between accuracy and sensitivity, as well as computational effectiveness.

1.3 Proposed Solution

This study introduces the EMLP-AGTO algorithm to tackle these obstacles. The main characteristics of the solution consist of:

- **Improved Pre-Processing:** Employing the Wiener filter to minimize noise as well as improve image quality, thereby facilitating superior feature extraction from biomedical images.
- **Advanced Feature Extraction:** The LTrP method is utilized to extract spatial as well as texture features essential for precise lung cancer classification.
- **Optimized Classification Framework:** The EMLP-AGTO algorithm fine-tunes MLP parameters through an innovative gorilla-motivated metaheuristic method, greatly improving computational effectiveness as well as maximizing precision.

The suggested method seeks to offer a scalable and effective, as well as accurate, solution for lung cancer detection,

tackling the existing challenges of precision, image quality, and optimization effectiveness in biomedical image investigation.

The organization of the paper is as follows. Section 1 is the introduction of biomedical image analysis for the detection of lung cancer. Section 2 is a literature survey. Section 3 introduces methodology with a proposed model, data collection, pre-processing by Wiener filter, feature extraction by LTrP, detection by EMLP, and AGTO algorithm. Section 4 is results and analysis. Section 5 is the conclusion.

2 Related Work

In order to partition the biological image, a deep guiding framework was implemented (Yin et al. 2020). A guided image filter module in the suggested framework restored the structural data via the guiding image. The approach allowed for quick inference (43 ms for a single image) and end-to-end training. Comprehensive investigations were done in cup and optic disc segmentation as well as vascular segmentation. The technique's efficacy was confirmed by four publicly accessible dataset experiments: DRIVE, REFUGE, ORIGA, and CHASEDB1.

In this work, a novel Hybrid Metaheuristics with Deep Learning based Fusion Model Biomedical Image Analysis (HMDL-MFMBIA) approach was described (Obayya et al. 2023). Initially, Swin-UNet-oriented segmentation as well as image pre-processing were carried out via the HMDL-MFMBIA approach. Moreover, Xception as well as the Residual Network (ResNet) method was used to combine many DL-oriented feature extractors. Lastly, bio-medical image identification as well as classification may be accomplished through the use of the Gated Recurrent Unit (GRU) method. To describe the outcomes of the approach, a large-scale simulation was carried out. The results related to the simulation suggested that the HMDL-MFMBIA method performed better than various DL methods.

The purpose of this work was to identify nodules by means of a computer-oriented visual process approach for size computation, utilizing recently created software (Nagihan Bilal et al. 2019). Using voice acoustic analysis, the impacts related to the nodule base as well as breadth ratios were assessed. Seventy-two cases of paediatric vocal nodules were evaluated. Nodules were detected utilizing the ImageJ News software. Automatic segmentation was employed. The results of this study identified new ratios in children with vocal nodules in collaboration with the biomedical engineering department.

The method was developed through inverse dynamics as well as video analysis techniques (Tianxing Gong et al. 2022). It was shown that in contrast to noninjurious skiing

motions, the maximum response force at the knee joint significantly increased during a dangerous skiing activity. The results were shown to be reliable as well as consistent with the conclusions of remaining researchers. In applications for preventing sports-related injuries, this method could serve as a more straightforward alternative to the complex analytical techniques currently employed.

To examine treatment plans, this study proposed a tiered approach utilizing Wrapper feature selection and Fuzzy C-Mean (FCM) clustering, as well as twelve classifiers (Ershadi and Rise 2022). Employing biomedical information, clinical information, as well as various imaging information, the proposed method evaluated the effectiveness linked to previous and current treatment strategies, applying a hierarchical decision-making approach to determine the best path forward. A case study was presented to illustrate the effectiveness of the proposed methodology. The dataset was analyzed using information preparation as well as the knowledge of specialists. Subsequently, the FCM clustering method was utilized to enhance classifier learning.

This research employed the modified cuckoo search method as well as a fuzzy system to address several challenges (Shouvik Chakraborty and Mali 2022). The proposed approach was evaluated using both quantitative and qualitative methods. The outcome was measured using standard evaluation metrics like SSIM, PSNR, MSE and Mean, as well as SD. The outcomes were compared with specific conventional techniques for each among the three objective functions as well as a diverse number of clusters. The proposed approach frequently outperformed its competitors and also produced significant improvements. These encouraging results motivated the implementation of the proposed scheme in real-world scenarios.

This technique was referred to as the Fuzzy Modified Cuckoo Search with Spatial Exploration (FMCSSSE) (Shouvik Chakraborty 2024). An important feature of image information was the strong correlation among pixels, with neighboring pixels showing similar feature information. To generate a significant segmented image, it was essential to consider the spatial information. Utilizing the traditional fuzzy clustering approach for leveraging spatial information was not suitable. Consequently, the fuzzy-modified cuckoo search method was employed to analyze spatial information as well as determine optimal clusters from biological images. This approach was applied to diverse biological images as well as compared with various advanced unsupervised techniques like MCS, FMCS, FEMO, and CS.

An extensive study on labeling guidelines and their impact on the quality of region annotations was presented (Tim Radsch et al. 2023). An examination of international competitions as well as professional practice revealed a variation in the needs for labelling instructions among

annotators, as well as in their current availability and quality. Lastly, expert annotators consistently outperformed crowd workers on Amazon Mechanical Turk. The necessity for biological image analysis was brought to light by this work.

This research's main contribution was the creation of a Big Data Architecture (BDA), which generated accurate predictions regarding patients' health problems as well as real-time alerts, therefore increasing the reliability of medical frameworks (Laila Almutairi et al. 2023). Comprehensive experiments were carried out on publicly available datasets. According to the simulation research, the recommended BDA-BMIC methodology surpassed other methods.

For a U-shaped symmetrical framework, a lightweight multi-convolutional Multi-Scale Convolutional Network block (MSConvNeXt) was created (Ma et al. 2024). It efficiently as well as economically captured semantic feature data. Additionally, a channel shuffling mechanism was suggested to encourage an effective feature fusion between the feature maps in a dynamic manner. The network block was now correctly installed within TriConvUNeXt, a U-shaped symmetrical encoder-decoder type framework. The suggested methodology was verified using a publicly available benchmark dataset, undergoing a thorough assessment in terms of segmentation effectiveness and computational expense in comparison to 13 baseline techniques. Specifically, TriConvUNeXt achieved 81% as well as a 97% reduced computational cost while obtaining a 1% higher Dice-Coefficient compared to TransUNet and UNet.

The examined research (Wang et al. 2022, 2023, 2024a, 2024b; Rajinikanth et al. 2022; Mukesh et al. 2022; Abdalredha Kadhim and Abed Mohammed 2024) emphasized several improvements in medical image analysis through AI methods. The UNet employing dual training emphasized lung segmentation in chest X-rays, showcasing enhanced accuracy for structural delineation, yet it did not possess the ability to categorize lung cancer sub-types utilizing high-resolution histopathological images. The hybrid CNN method integrated CNNs with swarm intelligence for classifying lung diseases, yet it failed to specifically tackle lung cancer subtypes and lacked adequate hyperparameter tuning for improved effectiveness. The analysis of AI in diagnosing kidney cancer highlighted the necessity for strong algorithms to manage imaging inconsistencies yet failed to apply these findings to lung cancer or provide a tailored approach for lung tissue examination. These studies together highlighted advancements yet showed a requirement for more targeted approaches in lung cancer categorization.

This study developed a novel method for identifying lung cancer through the use of Computed Tomography (CT) images (Karthika and Antonidoss 2026). Initially, the input images were collected as well as subsequently pre-processed

using a median filter. Additionally, the lung lobe was partitioned using the Modified U-Net architecture. After the segmentation was completed, feature extraction was carried out utilizing Maximally Stable Extremal Regions (MSER), Discrete Wavelet Transform (DWT), as well as geometric attributes. Finally, lung cancer identification was performed utilizing the Dense Residual Forward Taylor Network (Den-ResFT-Net). This method combined DenseNet, Deep Residual Network (DRN), as well as Taylor Series.

This study presented an advanced diagnostic framework that utilized Deep Learning (DL) with an optimization strategy as well as Explainable AI (XAI) to automate the interpretation of CT images for predicting lung cancer (Robert et al. 2025). To preserve contrast as well as edge details, the framework utilized a pre-processing technique on the basis of the Fast Local Laplacian Filter (FLLF). To ensure spatial consistency, Fuzzy C-Means (FCM) clustering integrated with Markov Random Fields (MRF) was employed for segmentation, which divided the pre-processed image, enhancing the prediction accuracy of lung cancer. Contour shape descriptors were used to extract morphological details from segmented areas to offer a strong characterization of lung cancer forms. Subsequently, these traits were categorized using the Pyramid Vision Transformer (PVT)–ResNet method, which improved classification precision by combining the local feature extraction capabilities of Attention Residual as well as Convolution Neural Network (CNN) with the global attention procedure of PVT.

The detected research gaps highlighted the insufficient emphasis on specific lung cancer subtypes as well as the limitation of sophisticated optimization methods designed for accurate histopathological image evaluation. Numerous current approaches fail to incorporate high-resolution image features essential for lung cancer diagnosis, while some concentrate on general lung conditions or various cancers, such as kidney cancer, without tackling lung cancer-specific issues. The suggested EMLP-AGTO tackles these issues by utilizing the AGTO for efficient parameter optimization, concentrating on histopathological classification of lung cancer subtypes, as well as incorporating enhanced texture extraction via LTrP to improve method precision as well as resilience in diverse medical imaging contexts.

In summary, although previous research in lung cancer histopathology classification has shown the efficacy of manually created features and hybrid learning frameworks, as well as deep CNN methods, numerous unresolved issues persist. Numerous conventional techniques have difficulty recognizing fine texture differences present in H&E slides, while deep learning methods typically demand significant computational power as well as vast annotated datasets to prevent overfitting. Moreover, only a limited number of studies have investigated lightweight methodologies that

can achieve high accuracy with minimal inference delay, which is essential for incorporation into standard clinical processes. These constraints collectively emphasize the necessity for a strategy that merges strong feature extraction with computational effectiveness as well as reliable optimization. Driven by these gaps, the suggested EMLP-AGTO methodology aims to balance discriminative strength with practical usability, employing LTrP descriptors as well as AGTO-based parameter adjustment to ensure consistent effectiveness with a lesser computational load.

2.1 Research Gaps

While automatic systems can quickly identify hidden patterns or correlations within biological image information, humans might not be able to do so. It could be able to identify earlier missed subtle characteristics or connections utilizing deep learning and data mining algorithms, which will help with diagnosis and medical research. Traditional human-oriented diagnostic methods are laborious and subject to various errors. Thus, automated healthcare environments incorporate biological image analysis. Computer vision may sometimes be superior to human vision. When applied to the study of biological images, it was more precise and accurate in identifying and deciphering minute features as well as abnormalities in biological images than human observers are. Nevertheless, image quality is critical for any image processing method. The image evaluation technique's output cannot reach the necessary accuracy if the image quality is poor. There are times when the processing methods take too long, which makes them unsuitable for practical use. Biomedical image analysis processing methods are sometimes hindered by time restrictions, which reduces their applicability in clinical settings. For various processing approaches, the effort and time required to perform their computations and complex algorithms make them excessively ineffective. By resolving these time-associated issues, processing approaches may be applied in medicine much more effectively. In order to overcome these challenges and make it possible for these techniques to be applied in real-world medical contexts, scientists are looking at ways to increase the effectiveness of algorithms, employ hardware acceleration, process data in parallel, and create real-time processing structures.

3 Proposed Methodology

The proposed biomedical image analysis model for lung cancer detection involves a multi-step approach designed to maximize accuracy, efficiency, and reliability. The method combines advanced image processing, feature extraction,

and classification techniques optimized through a novel metaheuristic algorithm. The main steps include data collection, image pre-processing, feature extraction, and classification using the Enhanced Multi-Layer Perceptron-based Artificial Gorilla Troops Optimizer (EMLP-AGTO). Every step is described thoroughly below:

Data Collection The method employs the LC25000 dataset, which serves as a standard benchmark for analyzing lung cancer images. This dataset comprises three categories of images: benign lung tissue, lung adenocarcinoma and squamous cell carcinoma of the lung. Every class has 3000 instances, offering a thorough data set for training as well as evaluating the method. Utilizing a benchmark dataset guarantees that the method's effectiveness can be matched against various advanced techniques.

Image Pre-Processing A Wiener filter is utilized to improve the quality of the images. The Wiener filter aims to diminish noise as well as improve the overall quality of the input images. It operates by reducing the MSE between the estimated and original images, enabling clearer and more accurate information for use in the following phases. The filter considers the statistical characteristics of noise in the images as well as utilizes a linear estimation to generate a de-noised image, essential for enhancing feature extraction precision.

Feature Extraction Using LTrP Following pre-processing, the images are subjected to feature extraction via the LTrP technique. This method gathers spatial as well as texture characteristics by examining the connection among the central pixel and its surrounding pixels in both horizontal and vertical orientations. The LTrP computes the first as well as second-order derivatives, resulting in a collection of 13 binary patterns (which include derivative magnitudes). These patterns are utilized to create histograms that reflect the structural as well as textural characteristics of lung tissues, essential for distinguishing among various kinds of lung cancer. The technique's capacity to capture detailed texture data guarantees that even minor differences in the images, vital for precise classification, are well described.

Classification Using EMLP-AGTO The extracted features are fed into a classification method utilizing an EMLP architecture. This method is optimized by leveraging the AGTO algorithm to adjust the MLP parameters for peak effectiveness.

- **MLP Structure:** The EMLP is made up of input and hidden, as well as output layers. The concealed layers consist of neurons utilizing activation functions (ReLU,

Tanh) that aid in modeling the non-linear connections among input features as well as output categories.

- **AGTO Optimization:** The AGTO algorithm, inspired by the social and adaptive behavior of gorillas, optimizes the MLP parameters. It consists of two primary stages:
 - **Exploration Stage:** Gorillas migrate to known and unknown locations, emulating exploration behaviors in the search for optimal solutions.
 - **Development Stage:** Gorillas follow or compete with other members to refine their positioning, representing the refinement of hyperparameters for maximizing accuracy.

The AGTO algorithm employs equations as well as parameters (e.g., migration functions, random counts) to iteratively refine the MLP format, with the goal of maximizing the

objective function—accuracy in this scenario. The outcome describes a finely adjusted EMLP method that effectively categorizes lung cancer images into the three specified groups.

Performance Evaluation The EMLP-AGTO method is assessed in comparison to multiple traditional approaches like HMDL-MFMBIA, FCM, and BDA-BMIC, as well as MSConvNeXt, using different performance measures involving accuracy, precision, sensitivity, F1 Score, and MCC.

The approach is depicted in Fig. 1, illustrating the progression from information gathering and image enhancement with the Wiener filter to feature extraction through LTrP, as well as classification utilizing the EMLP-AGTO

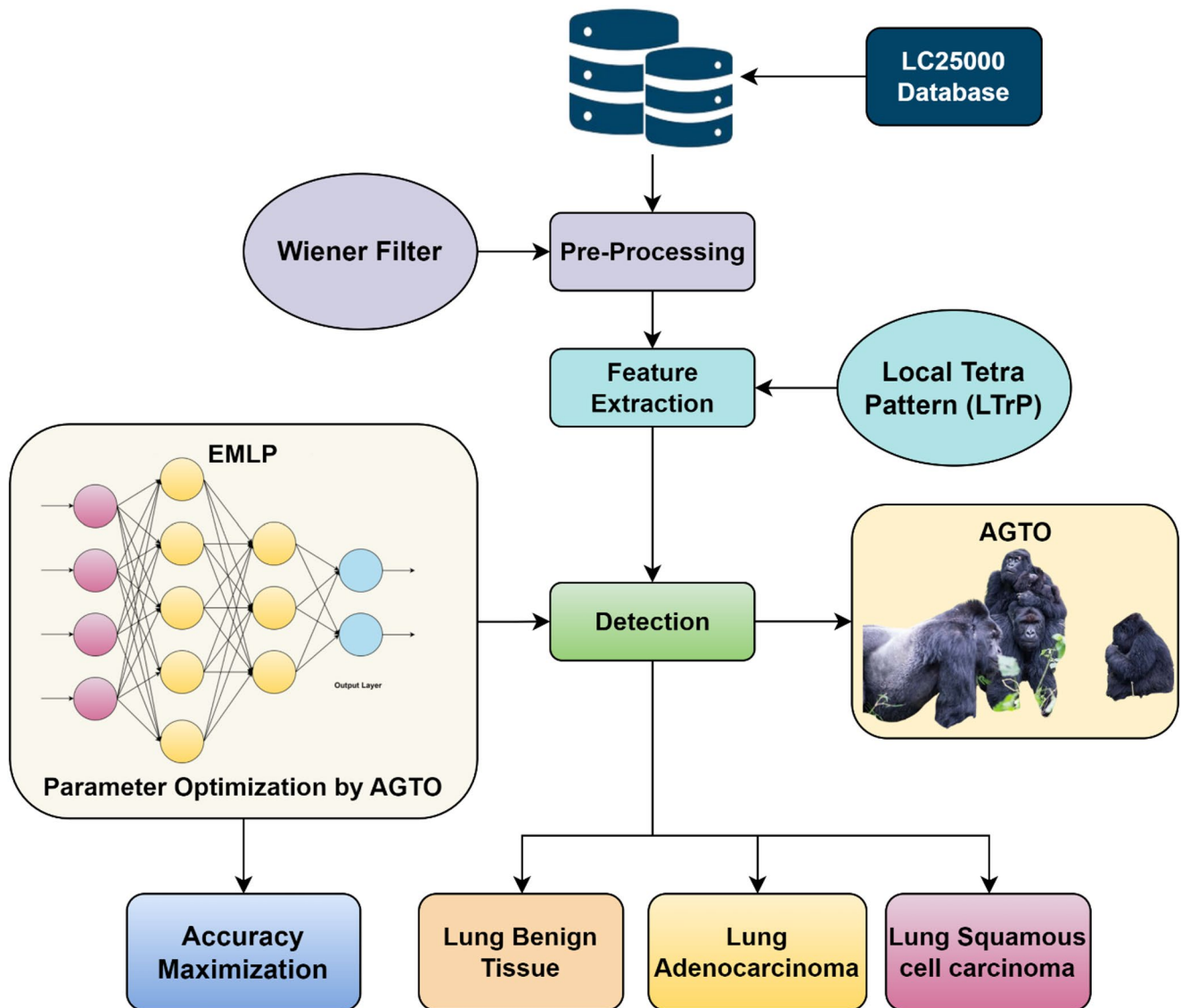


Fig. 1 Proposed biomedical image analysis model for the detection of lung cancer

algorithm. This graphical depiction aids in grasping the logical sequence as well as the connection of every element. The suggested approach integrates sophisticated image processing as well as optimization methods to develop a highly effective and precise method for lung cancer detection. Employing AGTO to optimize the MLP method guarantees improved classification accuracy alongside computational effectiveness, thus accomplishing this method as ideal for real-world clinical contexts in which prompt as well as dependable diagnoses are essential.

3.1 Proposed Model

The proposed biomedical image analysis model for the detection of lung cancer involves various steps like data collection, pre-processing, feature extraction, and detection. First, the information is taken from the LC25000 database, which is a common benchmark dataset. The Wiener filter is used in the recommended approach to pre-process

the input images. The proposed technique further employs an LTrP feature extractor to produce a collection of feature vectors. The EMLP classifier in combination with AGTO is used to detect lung benign tissue, lung adenocarcinoma, and lung squamous cell carcinoma, respectively. The proposed EMLP-AGTO-based biomedical image analysis model for the detection of lung cancer is diagrammatically shown in Fig. 1.

3.2 Data Collection

The data related to the proposed biomedical image analysis model is gathered from the standard benchmark dataset called the LC25000 database. It is composed of three classes with 3000 samples under every class. AI researchers get unrestricted access to de-identified, HIPAA-compliant, verified images at no cost. Some of the sample collected images are shown below in Fig. 2.

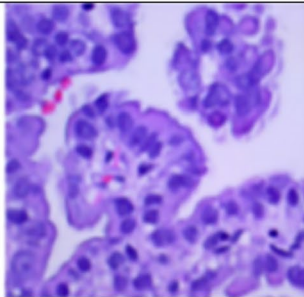
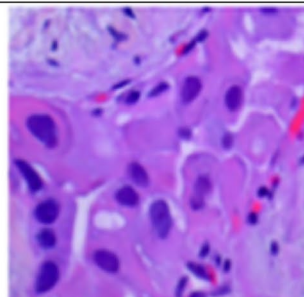
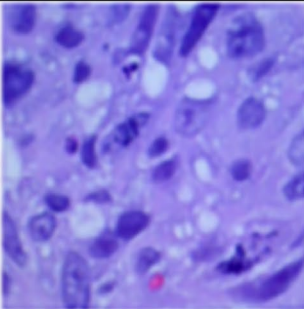
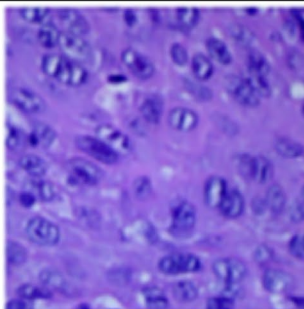
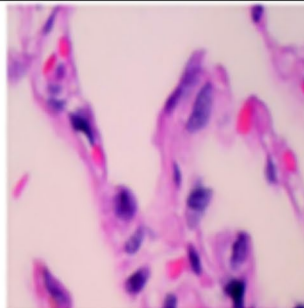
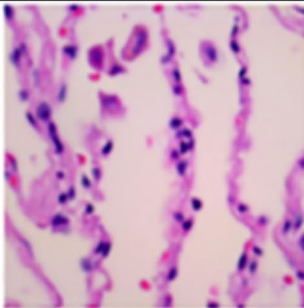
Image description (Classes)	Image 1	Image 2
Lung Benign Tissue		
Lung Adenocarcinoma		
Lung Squamous Cell Carcinoma		

Fig. 2 Sample collected images of the dataset

3.3 Pre-processing by Wiener Filter

The gathered images from the suggested biomedical image analysis method for lung cancer detection undergo a pre-processing stage executed with the Wiener filter. Additive noise greatly affects the inverse filtering process. It is carried out via Wiener filtering. The Wiener filter decreases the MSE between the desired process and the estimated process. It decreases the overall average squared error. The Wiener filter provides a linear estimation related to the original image. Wiener filters effectively decrease noise in images by assessing the image's noise level against an estimation of the desired signal without noise. It is based on statistical analysis. Three essential traits characterize Wiener filters.

Requirement The filter must be causal as well as physically implementable.

Assumption Linear stochastic processes that are stationary and involve both noise and image, either with known spectral features or with known cross correlation and autocorrelation.

Performance criterion This filter is frequently used in the deconvolution process.

The Wiener filter is commonly utilized for its speed and ease of use. The Fourier Domain Wiener Filter is given as below:

$$H(v, w) = \frac{I^*(v, w) Q_t(v, w)}{|I(v, w)|^2 Q_t(v, w) + Q_o(v, w)} \quad (1)$$

Dividing by Q_t describes the explanation of its characteristic:

$$H(v, w) = \frac{I^*(v, w)}{|I(v, w)|^2 + \frac{Q_o(v, w)}{Q_t(v, w)}} \quad (2)$$

Here, the PSD related to an un-degraded image is described by $Q_t(v, w)$, PSD related to noise is described by $Q_o(v, w)$, complex conjugate liked with the degradation function is shown by $I^*(v, w)$, and the degradation function is shown by $I(v, w)$ correspondingly.

The phrase $\frac{Q_o}{Q_t}$ may be considered as the SNR reciprocal. The intensity of noise in an image, or the variation in noise, dictates the functionality of this filter. The filter does not smooth much when the variation is big, but it smooths considerably when the variation is small.

3.4 Feature Extraction by LTrP

The pre-processed images of the biomedical image analysis model for the detection of lung cancer are given to the feature extraction process that is performed by the LTrP method for extracting the features. Utilizing first order derivatives in both horizontal and vertical directions, the LTrP approach calculates the directions that represent the connection between the central pixel and its neighbors. The direction related to the centre grey pixel is used by the LTrP to characterize the spatial structure associated with the local texture. The first order derivative at centre pixel h_d in this technique may be expressed as

$$J_{0^0}(h_d) = J(h_i) - J(h_d) \quad (3)$$

$$J_{90^0}(h_d) = J(h_w) - J(h_d) \quad (4)$$

When h_i and h_w represent, respectively, the horizontal as well as vertical neighbors of h_d , and the centre pixel's orientation may be computed as:

$$J_{dir}^1(h_d) = \begin{cases} 1, & J_{0^0}^1(h_d) \geq 0 \text{ and } J_{90^0}^1(h_d) \geq 0 \\ 2, & J_{0^0}^1(h_d) < 0 \text{ and } J_{90^0}^1(h_d) \geq 0 \\ 3, & J_{0^0}^1(h_d) < 0 \text{ and } J_{90^0}^1(h_d) < 0 \\ 4, & J_{0^0}^1(h_d) \geq 0 \text{ and } J_{90^0}^1(h_d) < 0 \end{cases} \quad (5)$$

It is clear from Eq. (5) that the image is translated into four values, 1, 2, 3, and 4, which may be thought of as four directions. The second order derivative (LTrP²) must then be computed in order to provide an 8-bit tetra pattern for every centre pixel. The orientation associated with the centre pixel is next used to split this tetra pattern into four halves. Lastly, three binary patterns are created from the tetra patterns for every direction. In other words, twelve (4×3) binary patterns are formed for four orientations if the direction associated with the centre pixel is 1. The 13th binary pattern (LP), which is computed utilizing the magnitudes of the vertical and horizontal first-order derivatives, is employed in the LTrP approach.

$$N_{J^1(h_q)} = \sqrt{(J_{0^0}^1(h_q))^2 + (J_{90^0}^1(h_q))^2} \quad (6)$$

$$LP = \sum_{q=1}^Q 2^{(q-1)} \times g_1(N_{J^1(h_q)} - N_{J^1(h_d)}) \Bigg|_{Q=8} \quad (7)$$

Following the identification related to the 13-bit binary pattern, their histogram is computed, and the query image is contrasted with the database's images. The O best images are chosen throughout the comparison procedure. When it

comes to retrieval, LTrP works better than LBP, LDT, and LTP and can extract more specific data from the image.

3.5 Detection by EMLP

The extracted features are subjected to the final detection phase that is accomplished employing the novel EMLP. Here, the parameters of MLP are tuned by AGTO with the aim of attaining accuracy maximization as the main fitness function, thus referred to as novel EMLP. A feed forward Neural Network (NN), or MLP, contains an output layer, one or multiple hidden layers, and at least one input layer, each of which performs a distinct function. A multitude of neurons having activation functions may be found in both the hidden and output layers. NN design may be carefully constructed to simulate complicated non-linear relationships among input factors as well as the output.

Because the input layer is not counted towards the total count of layers, a standard ANN is either a one-hidden-layer MLP or a two-layer NN. A 2-layer neural network is defined as a function $g : E \rightarrow M$, in which E represents the size associated with the input vector y and M denotes the size related to the output vector $g(y)$. Using weight matrices $X(1)$ and $X(2)$, activation functions H and h , bias vectors $c(1)$ and $c(2)$, and matrix notation,

$$g(y) = H\left(c^{(2)} + X^{(2)}\left(h\left(c^{(1)} + X^{(1)}y\right)\right)\right) \quad (8)$$

The learning procedure imposes needs on the activation function. It must be a continuous function that is bounded, non-constant, and increases monotonically. It is ensured by utilizing this function that the derivation is real and useful for network learning.

A real-valued number is sent through the Sigmoid function, which converts it into the range between 0 and 1. Generally speaking, big positive values become 1, and large negative numbers become 0. The sigmoid function has two drawbacks: it is not zero-centered and saturates during learning. A real-valued number is converted via the Tanh function to a range between -1 and 1 . Its activations saturate just similarly to those of a sigmoid neuron, but its output is zero-centered. For this reason, the Tanh is favored over the Sigmoid. The ReLu only thresholds negative values at zero and is non-saturating and linear. ReLu has the benefits of straightforward computation and a much faster rate of convergence than Tanh and Sigmoid. The drawback is that because of the big gradient update, ReLu may be turned off during learning. The final function given, Maxout, is significantly distinct due to the twofold ReLu function. This feature is not turned off when learning and has the advantages

of ReLu. The drawback is that it generates twice as many parameters, which need to be optimized.

Every parameter needs to be optimized for the net to be trained. To be learnt are the parameters group $= \{X(2), c(2), X(1), c(1)\}$. The backpropagation technique may be used to acquire the gradients, which are then used to update the weights. The gradient is calculated for each weight X and bias C in the network utilizing the loss function $(X, C|k)$, and it is transmitted from the output to the input. The loss function for NN classification is, for instance:

$$L(X, C|k) = \sum_{\hat{z} \in 0} z^{(k)} \log(\hat{z}^{(k)}) + \log(1 - \hat{z}^{(k)}) (1 - z^{(k)}) \quad (9)$$

Here, z describes the actual class and $\hat{y}$ describes the outputs from the output layer. NNs can be universal approximators having just one hidden layer, or they can have several layers with multiple units. This is only an actual finding.

The $M1$ and $M2$ regularizations represent the major prevalent. $M1$ denotes a variation in every weight x to $|x|$, in which $|x|$ describes the regularization strength. This results in sparser weight vectors during optimization, the usage of a sparser subset associated with the major crucial inputs, and almost invariant units to the noisy inputs.

$M2$ denotes that every weight w in the network is varied to $|x|$, in which regularization strength $|x|$ is once again present. This permits the network to utilize all of its inputs other than a subset of them extensively, strongly penalizing peak weights and favoring diffuse weight vectors. Combining these two regularizations describes an option. Typically, the hyper parameter represents a tiny value, like $0-3$. Lastly, the function to optimize for the outcome is:

$$L^1(X, C|k) = L(X, C|k) + \lambda_1 M1(X, C|k) + \lambda_2 M2(X, C|k) \quad (10)$$

Another highly powerful regularization strategy represents the Dropout. Dropout is achieved during training by either placing a neuron to zero or merely maintaining it active with a probability q . The value of the hyper parameter q , for instance, is 0.2 and ranges from 0 to 1 .

$$g[n, o] * h[n, o] = \sum_{k=1}^K \sum_{j=1}^J g[j, k] h[n - j, o - k] \quad (11)$$

The traditional MLP has various advantages, such as handling complex nonlinear problems, performing faster predictions after training, managing a vast quantity of input data, etc. But it suffers from a non-convex loss function, sensitivity to feature scaling, computational complexity problems, etc. Thus, to overcome its drawbacks, the parameters

of MLP are tuned by the AGTO algorithm with the aim of deriving the accuracy maximization as the fitness function, thus called the novel EMLP. This novel EMLP eradicates the computational complexity problem and also overcomes loss function, etc.

3.6 AGTO Algorithm

The AGTO algorithm is selected here in the proposed biomedical image analysis model for the detection of lung cancer for optimizing the parameters of the ELM method in order to derive the accuracy maximization as the objective function. The suggested population intelligence optimization algorithm, known as the AGTO, is dependent on the social behavior as well as the collective lifestyle of gorillas. A group of silver-backed or adult male gorillas, along with numerous adult female gorillas as well as their offspring, make up the “troops” of gorillas that inhabit the forest. The silver-backed gorilla serves as the group’s leader, settling conflicts, deciding on group activities, assisting the gorillas in locating food sources, and overseeing the group’s security as well as well-being.

The AGTO algorithm simulates the group behavior of gorillas using five distinct operators, which are primarily separated into two stages: the exploration stage as well as the development stage. Three distinct mechanisms—migration to known areas, migration to other gorilla positions, and migration to unknown locations—were used throughout the exploration stage. Two social behaviors were picked up during the developmental stage: competing with adult female gorillas as well as trailing the silver-backed gorilla. The gorillas’ location update during the exploring stage is displayed in Eqs. (12), (13), and (14):

$$HY(u+1) = (UB - LB) \times s_1 + LB, \text{ Rand} < t \quad (12)$$

$$HY(u+1) = (s_2 - D) \times Y_s(u) + M \times I, \text{ Rand} \geq 0.5 \quad (13)$$

$$HY(u+1) = Y(u) - M \times (M \times Y(u) - HY_s(u)) + s_3 \times (Y(u) - HY_s(u)), \text{ Rand} < 0.5 \quad (14)$$

$Y(u)$ represents the gorilla’s present location, and $HY(u+1)$ describes the potential location for the gorilla in the following iteration. Moreover, the random numbers Rand , s_1 , s_2 , and s_3 have values of (0,1). Prior to optimizing operations, a parameter called t has to be assigned a value. Typically, this value is 0.03. This option establishes the likelihood that the gorilla will migrate to an unidentified area. The variables’ upper as well as lower bounds are denoted by the symbols UB and LB , accordingly. Individual gorilla locations Y_s and HY_s are chosen at random

from a population. The formulae (15), (16), and (17), correspondingly, reflect the parameters D , M , and I :

$$D = (\cos(2 \times s_4) + 1) \times \left(1 - \frac{it}{\max it}\right) \quad (15)$$

$$M = D \times m \quad (16)$$

$$I = A \times Y(u) \quad (17)$$

The present iteration value describes the parameter of it in Eq. (15), and the overall iteration value for carrying out the optimization procedure is $\max it$. s_4 describes a random number that ranges from 0 to 1. The parameter of m in Eq. (16) has a random value of $(-1, 1)$. A ’s parameter in Eq. (17) describes a random number that falls between $[-D, D]$ in the issue dimension. Gorillas select to use the silver backed gorilla procedure throughout the developmental stage; this behavior is represented in Eq. (18):

$$HY(u+1) = M \times N \times (Y(u) - Y_{silverback}) + Y(u) \quad (18)$$

$$N = \left(\left| \frac{1}{O} \times \sum_{j=1}^O HY_j(u) \right|^h \right)^{\frac{1}{h}} \quad (19)$$

$$h = 2^M \quad (20)$$

The location related to the silver-backed gorilla (optimal solution) is represented by $Y_{silverback}$ in Eq. (18), and the parameter of M describes a parameter that has to be assigned a value prior to the optimization operation in order to change the development stage procedure. The parameter of N is described by Eq. (19), in which h is described by Eq. (20), the parameter of O describes the total count of gorillas, and $HY_j(u)$ indicates the location associated with the j^{th} candidate gorilla at the u^{th} iteration. Gorillas prefer to compete with adult female gorilla procedures during the development stage; Eq. (21) illustrates their behavior.

$$HY(u+1) = Y_{silverback} - (Y_{silverback} \times R - Y(u) \times R) \times B \quad (21)$$

$$R = 2 \times s_5 - 1 \quad (22)$$

$$B = \beta \times F \quad (23)$$

$$F = \begin{cases} O_1, & \text{Rand} \geq 0.5 \\ O_2, & \text{Rand} < 0.5 \end{cases} \quad (24)$$

The impact force, described by Eq. (22), is simulated in Eq. (21) as the parameter of R , and the coefficient of

violence in the fight, given by Eq. (23), is simulated by the parameter of B . Equation (22), where s_5 describes a random number between 0 and 1, and Eq. (23), where β describes the parameter value supplied before to the optimization process. Equation (24), which represents the dimensions associated with the solution, is simulated by means of the parameter F . In the problem dimension, the parameter of F will be equal to the random value and the normal distribution if $Rand = 0.5$. On the other hand, the random value in the normal distribution will equal the parameter of F . The pseudo code of AGTO is shown in Algorithm 1.

4 Results and Analysis

To ensure a fair and representative evaluation, the proposed EMLP-AGTO model was benchmarked against four methods commonly applied in histopathology and medical image classification tasks. HMDL-MFMBIA was included as a hybrid feature-learning framework that combines hand-crafted descriptors with machine-learning classifiers, making it suitable for texture-rich H&E images. Fuzzy C-Means (FCM) represents a widely used unsupervised clustering technique that serves as a classical baseline for tissue differentiation. BDA-BMIC was chosen due to its integration

```

1:  Start
2:  Input the population size  $O$  and maximum iteration count  $U$  and parameters  $\beta$  and  $q$ 
    [extracted features of the proposed biomedical image analysis model for the detection of
    lung cancer]
3:  Output the Gorilla location as well as its fitness value [best solution with maximized
    accuracy of the proposed model]
4:  Random population initialization  $Y_j$  ( $j = 1, 2, \dots, O$ )
5:  Compute the fitness values of Gorilla [accuracy maximization of the proposed model]
6:  While (stopping criteria not attained) do
7:      
$$D = (\cos(2 \times s_4) + 1) \times \left(1 - \frac{it}{\max it}\right)$$

8:      
$$M = D \times m$$

9:      For (every Gorilla ( $Y_j$ )) do
10:          $HY(u + 1) = (UB - LB) \times s_1 + LB, Rand < t$ 
11:          $HY(u + 1) = (s_2 - D) \times Y_s(u) + M \times I, Rand \geq 0.5$ 
12:          $HY(u + 1) = Y(u) - M \times (M \times Y(u) - HY_s(u)) + s_3$ 
            
$$\times (Y(u) - HY_s(u)), Rand < 0.5$$

13:      End for
14:      Compute the fitness values of Gorilla [accuracy maximization of the proposed model]
15:      If  $HY$  is better than  $Y$ , replace them
16:      For (each Gorilla ( $Y_j$ )) do
17:         If ( $|D| \geq 1$ ) then
18:             $HY(u + 1) = M \times N \times (Y(u) - Y_{silverback}) + Y(u)$ 
19:         else
20:             $HY(u + 1) = Y_{silverback} - (Y_{silverback} \times R - Y(u) \times R) \times B$ 
21:         End if
22:      End for
23:      Compute the fitness values of Gorilla [accuracy maximization of the proposed model]
24:      If new solutions are advanced than earlier solutions, substitute them
25:      Place  $Y_{silverback}$  as the silverback location (optimal location) [best solution of the
    proposed model]
26:  End while
27:  Return  $Y_{best\ gorilla}$ , best fitness [optimal solution with maximized accuracy of the
    proposed model]
28:  Stop

```

Algorithm 1 AGTO

of improved feature extraction as well as decision-making techniques, providing a solid conventional machine-learning alternative. MSConvNeXt, a newly developed deep convolutional method featuring multi-scale processing abilities, was selected to exemplify contemporary deep learning architectures recognized for their excellent effectiveness on medical image datasets. Together, these approaches offer a thorough range of baselines across classical, and hybrid, as well as deep learning frameworks, facilitating a significant evaluation of the suggested method's effectiveness and precision.

For this research, the LC25000 dataset was randomly divided into 80% for training as well as 20% for testing, ensuring an equal description of classes using stratified sampling. This guaranteed that both malignant and benign instances were evenly distributed throughout the two groups. To preserve the authenticity of the handcrafted LTrP texture descriptors as well as prevent the introduction of artificial structural changes, no geometric or intensity-oriented information augmentation methods were utilized. Robustness was achieved by employing five-fold cross-validation, with every fold going via the entire pre-processing as well as the AGTO-MLP training process. This strategy provided a more reliable assessment of generalization without altering the inherent morphology of histopathology images.

4.1 Experimental Setup

The proposed novel EMLP-AGTO for the biomedical image analysis model for the detection of lung cancer was implemented in MATLAB, and the findings were analyzed. The population size and the iteration count were placed to be 10 and 100, respectively. The proposed EMLP-AGTO model was compared with distinct conventional methods like HMDL-MFMBIA, FCM, BDA-BMIC, and MSConvNeXt in terms of analysis like accuracy, precision, sensitivity, F1 Score, specificity, True Positive Rate (TPR), Matthew's Correlation Coefficient (MCC), and False Positive Rate (FPR) to prove the betterment of the proposed model. The parameters utilized in the proposed model and existing models have been elaborated in Table 1.

4.2 Accuracy Analysis

The accuracy analysis of the proposed EMLP-AGTO and the conventional methods for the biomedical image analysis model for the detection of lung cancer is displayed here in Fig. 3. It may be seen clearly that the suggested EMLP-AGTO beats the other techniques in all the iterations for the proposed model in terms of accuracy. The greater the accuracy, the better is the introduced EMLP-AGTO method. The proposed EMLP-AGTO method, in terms of accuracy, is 6.76%, 5.01%, 5.88%, and 3.54% better than HMDL-MFMBIA, FCM, BDA-BMIC, and MSConvNeXt,

Table 1 Algorithm and parameter specifications

Algorithm	Parameters
EMLP-AGTO (Proposed)	- Population Size: 10 - Number of Iterations: 100 - Learning Rate: Optimized via AGTO - Activation Functions: ReLU, Tanh, Sigmoid - Hidden Layers: Tuned based on AGTO optimization - Fitness Function: Accuracy maximization (AGTO)
HMDL-MFMBIA	- Segmentation Threshold: Determined via Swin-UNet module - Metaheuristic Parameters: Set for combining GRU layers and Residual Network - Learning Rate: Predefined and fixed across iterations
FCM	- Fusion Model Parameters: Configured for Xception and ResNet feature extractors - Number of Clusters: 5 (can vary based on dataset characteristics) - Fuzziness Coefficient (m): Typically set to 2 - Max Iterations: 100
BDA-BMIC	- Termination Criterion: Threshold set to determine convergence of clusters - Architecture Depth: Number of layers for deep learning fusion - Clustering Iterations: Set based on dataset volume - Gradient Descent Learning Rate: Fixed parameter to optimize classification
MSConvNeXt	- Big Data Parameters: Configurations for handling and segmenting large datasets - Number of Convolutional Layers: 15 - Filter Size: 3×3 kernels for convolution operations - Activation Function: ReLU - Optimizer: Adam with a learning rate set at 0.001 - Channel Shuffling: Parameters for efficient feature fusion

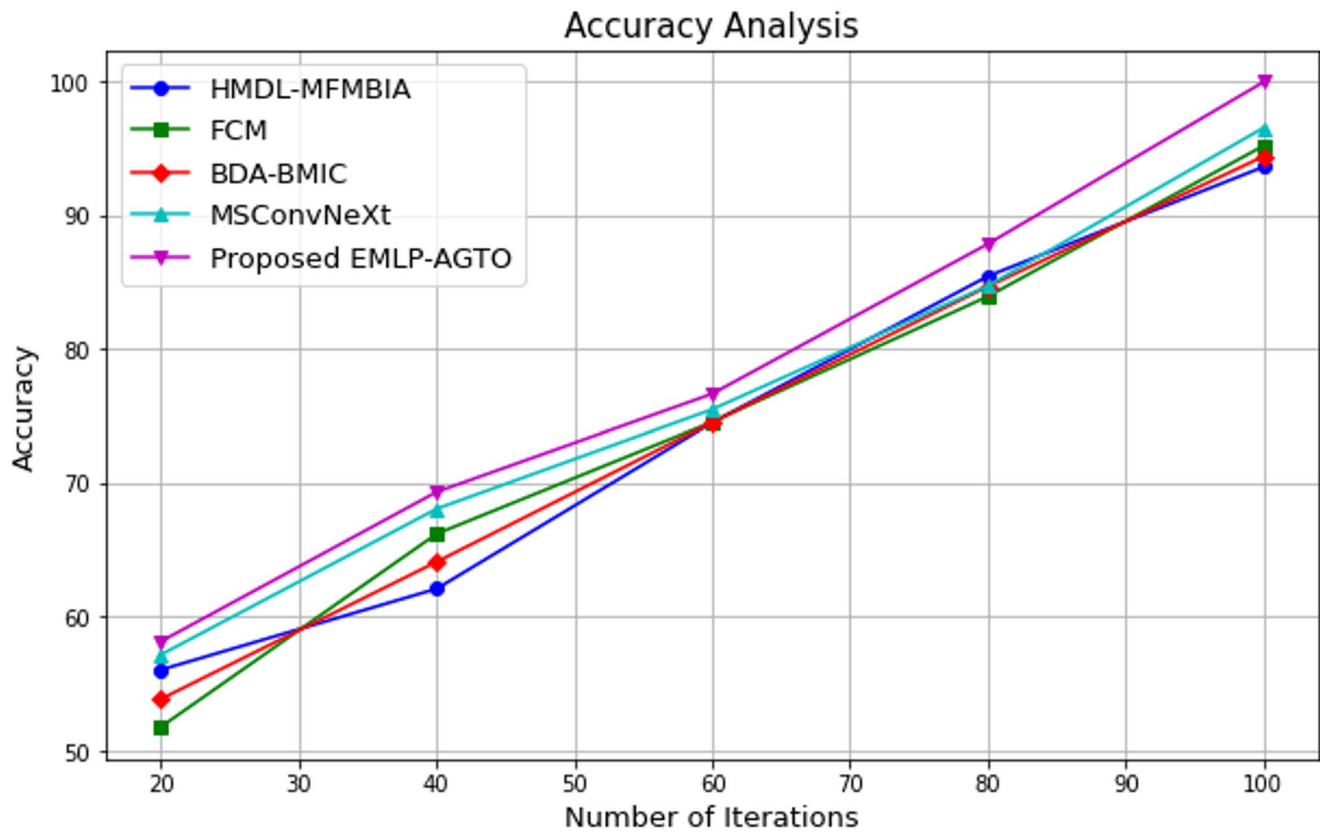


Fig. 3 Accuracy curve depicting convergence behavior across training epochs

Table 2 Accuracy analysis

Methods	Iterations				
	20	40	60	80	100
HMDL-MFMBIA (Obayya et al. 2023)	56.06	62.11	74.51	85.43	93.60
FCM (Ershadi and Rise 2022)	51.80	66.21	74.55	83.92	95.17
BDA-BMIC (Almutairi et al. 2023)	53.88	64.12	74.52	84.66	94.38
MSConvNeXt (Ma et al. 2024)	57.19	68.08	75.49	84.72	96.51
Proposed EMLP-AGTO	58.18	69.34	76.65	87.84	99.93

respectively. Hence, it can be demonstrated that the developed EMLP-AGTO surpasses the existing methods with respect to accuracy. Table 2 summarizes the accuracy analysis of the proposed system.

4.3 Precision Analysis

Figure 4 displays the precision analysis of the suggested EMLP-AGTO and the traditional approaches for the biomedical image analysis model for the diagnosis of lung cancer. It is evident that in every iteration of the proposed model, the suggested EMLP-AGTO performs more precisely than the other techniques. The introduced EMLP-AGTO approach

for biomedical image analysis for lung cancer diagnosis is better the higher the precision. The proposed EMLP-AGTO method with respect to precision is 5.81%, 4.03%, 4.92%, and 2.56% more advanced than HMDL-MFMBIA, FCM, BDA-BMIC, and MSConvNeXt, respectively. Therefore, it can be shown that the created EMLP-AGTO outperforms the current approaches for the biomedical image analysis model in terms of precision for lung cancer diagnosis. Table 3 summarizes the precision analysis of the proposed system.

4.4 Sensitivity Analysis

The sensitivity analysis of the conventional methods and the recommended EMLP-AGTO for the biomedical image analysis model for lung cancer detection is displayed in Fig. 5. It is clear that the recommended EMLP-AGTO outperforms the other methods in every iteration of the proposed model. The more sensitive the EMLP-AGTO technique, which was just established, is for biomedical image processing for lung cancer diagnosis, the better. The proposed EMLP-AGTO method, in terms of sensitivity, is 5.87%, 4.07%, 4.97%, and 2.58% higher than HMDL-MFMBIA, FCM, BDA-BMIC, and MSConvNeXt, respectively. Consequently, it can be demonstrated that in terms of sensitivity for lung cancer

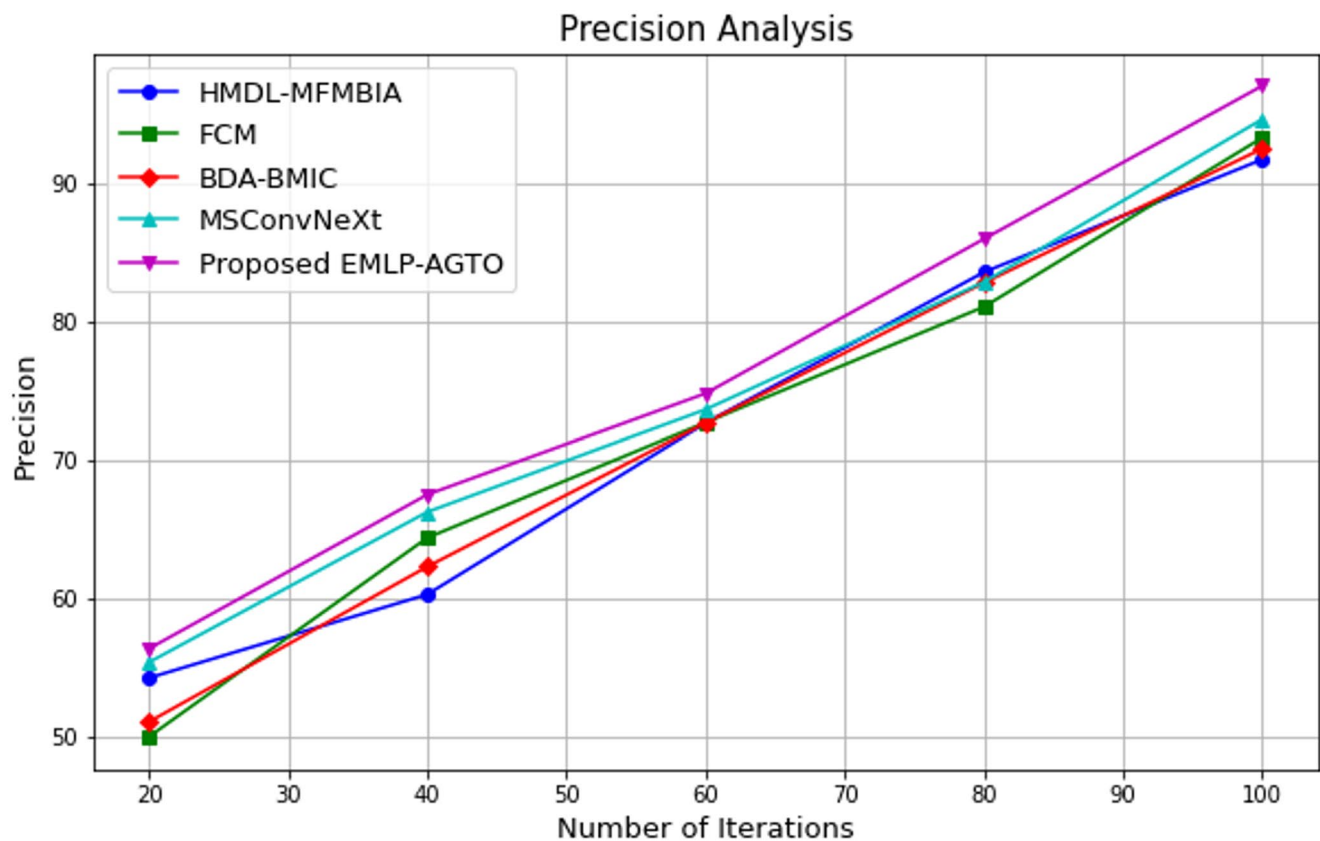


Fig. 4 Precision curve demonstrating the model's consistency in correctly identifying malignant cases

Table 3 Precision analysis

Methods	Iterations				
	20	40	60	80	100
HMDL-MFMBIA (Obayya et al. 2023)	54.19	60.24	72.64	83.56	91.73
FCM (Ershadi and Rise 2022)	49.93	64.34	72.68	81.05	93.30
BDA-BMIC (Almutairi et al. 2023)	51.01	62.25	72.65	82.79	92.51
MSConvNeXt (Ma et al. 2024)	55.32	66.21	73.62	82.85	94.64
Proposed EMLP-AGTO	56.31	67.47	74.78	85.97	97.06

detection, the developed EMLP-AGTO performs better than the existing methods for the biomedical image analysis model. Table 4 summarizes the sensitivity analysis of the proposed system.

4.5 F1 Score Analysis

Table 5; Fig. 6 show the F1 Score analysis of the suggested EMLP-AGTO and the standard approaches for the biomedical image analysis model for lung cancer diagnosis. In each iteration of the proposed model, it is evident that the suggested EMLP-AGTO performs better than the other techniques. The recently developed EMLP-AGTO approach

has a higher F1 Score and is intended for use in biomedical image processing to diagnose lung cancer. The proposed EMLP-AGTO method with respect to F1 Score is 6.13%, 4.25%, 2.84%, and 1.56% greater than HMDL-MFMBIA, FCM, BDA-BMIC, and MSConvNeXt, respectively. Thus, it can be shown that the proposed EMLP-AGTO outperforms the traditional approaches for the biomedical image analysis model in terms of F1 Score for lung cancer diagnosis.

4.6 Specificity Analysis

The specificity study of the recommended EMLP-AGTO and the conventional methods for the biomedical image analysis model for lung cancer detection are displayed in Table 6; Fig. 7. It is evident that the recommended EMLP-AGTO beats the other methods in every iteration of the proposed biomedical image analysis model for lung cancer detection. With greater specificity, the newly created EMLP-AGTO method is meant to be used in biomedical image processing for the diagnosis of lung cancer. The proposed EMLP-AGTO method, in terms of specificity, is 8.82%, 5.56%, 4.09%, and 3.93% superior to HMDL-MFMBIA, FCM, BDA-BMIC, and MSConvNeXt, respectively. Thus, it can be demonstrated that in terms of specificity for lung cancer detection, the recommended EMLP-AGTO accomplishes

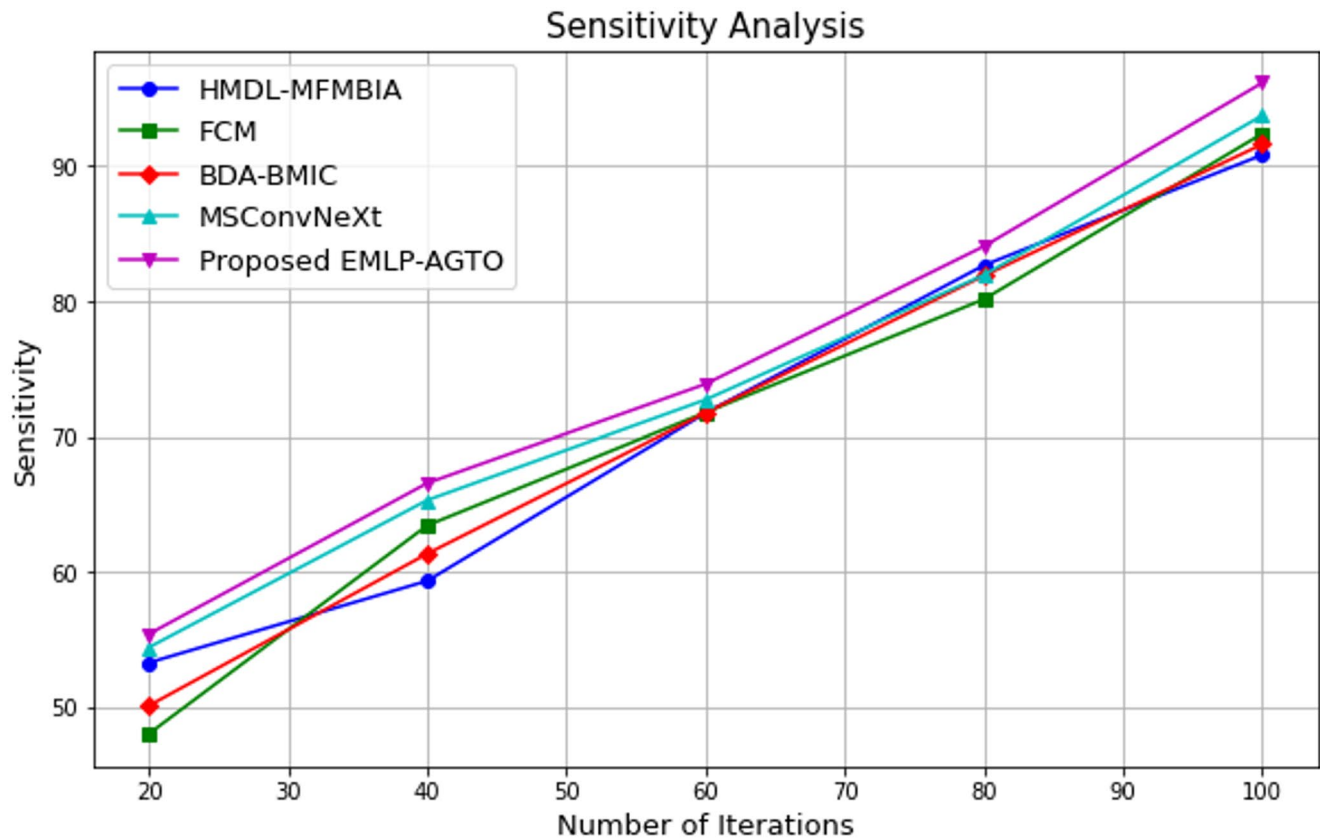


Fig. 5 Sensitivity (Recall) curve capturing the ability of the model to detect true malignant instances

Table 4 Sensitivity analysis

Methods	Iterations				
	20	40	60	80	100
HMDL-MFMBIA (Obayya et al. 2023)	53.30	59.35	71.75	82.67	90.84
FCM (Ershadi and Rise 2022)	48.04	63.45	71.79	80.16	92.41
BDA-BMIC (Almutairi et al. 2023)	50.12	61.36	71.76	81.90	91.62
MSConvNeXt (Ma et al. 2024)	54.43	65.32	72.73	81.96	93.75
Proposed EMLP-AGTO	55.42	66.58	73.89	84.08	96.17

Table 5 F1 Score analysis

Methods	Iterations				
	20	40	60	80	100
HMDL-MFMBIA (Obayya et al. 2023)	49.44	55.49	67.89	78.81	86.98
FCM (Ershadi and Rise 2022)	44.18	59.59	67.93	76.30	88.55
BDA-BMIC (Almutairi et al. 2023)	46.26	57.50	67.90	77.04	89.76
MSConvNeXt (Ma et al. 2024)	50.57	61.46	68.87	77.10	90.89
Proposed EMLP-AGTO	51.56	62.72	69.03	80.22	92.31

more than the conventional methods for the biomedical image analysis model.

4.7 TPR Analysis

Table 7; Fig. 8 show the TPR research of the suggested EMLP-AGTO and the traditional approaches for the biomedical image analysis model for lung cancer diagnosis. It is clear that in each iteration of the suggested biomedical image analysis model for lung cancer diagnosis, the suggested EMLP-AGTO outperforms the other techniques. The recently developed EMLP-AGTO approach, which has a higher TPR, is intended for application in biomedical image processing for lung cancer diagnosis. The proposed EMLP-AGTO method with respect to TPR is 10.72%, 9.04%, 6.05%, and 5.32% greater than HMDL-MFMBIA, FCM, BDA-BMIC, and MSConvNeXt, respectively. Thus, it can be shown that the suggested EMLP-AGTO performs better than the traditional approaches for the biomedical image analysis model in terms of TPR for lung cancer diagnosis.

The MCC research of the recommended EMLP-AGTO and the conventional methods for the biomedical image analysis model for lung cancer detection are displayed in Table 8; Fig. 9. It is evident that the recommended EMLP-AGTO performs better than the other methods in every

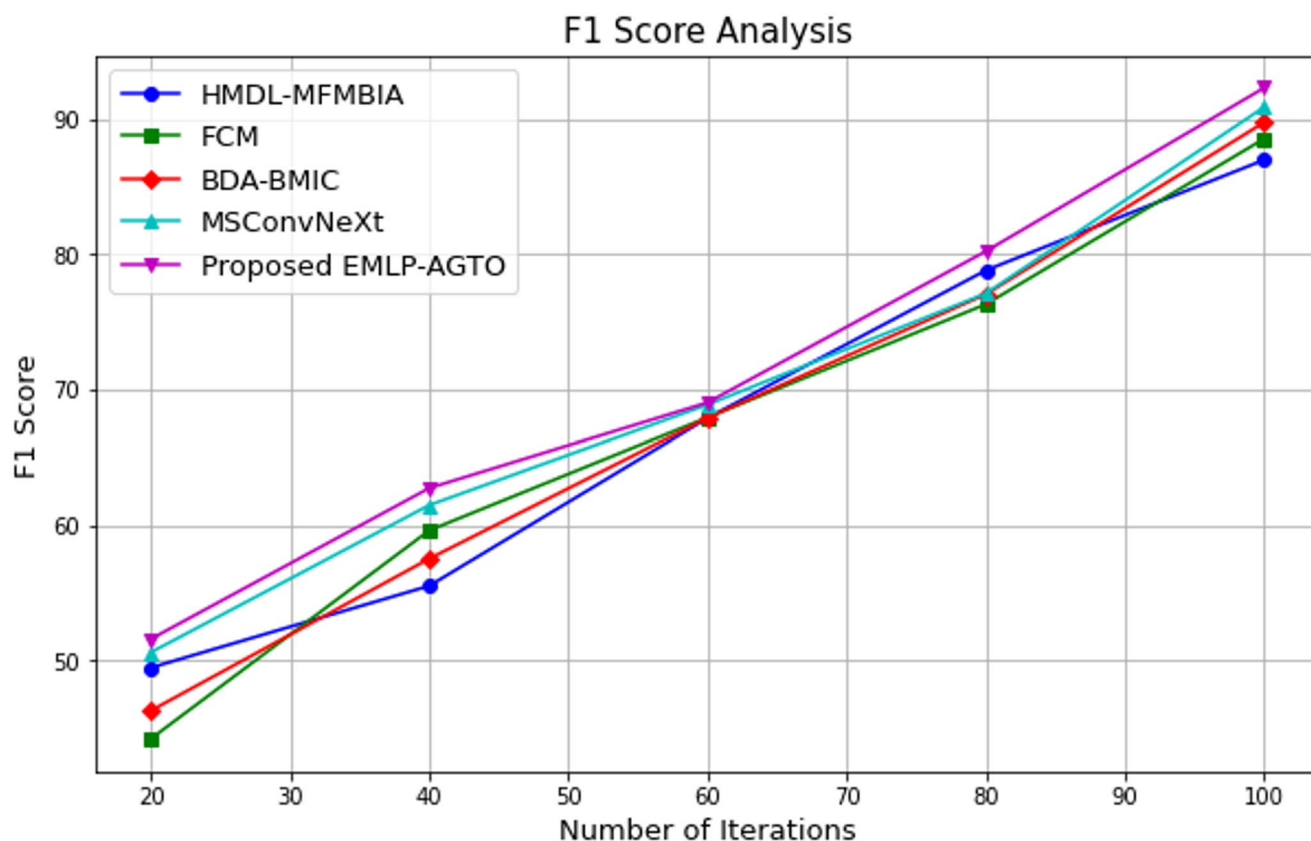


Fig. 6 F1-score progression highlighting the harmonic balance between precision and sensitivity

Table 6 Specificity analysis

Methods	Iterations				
	20	40	60	80	100
HMDL-MFMBIA (Obayya et al. 2023)	46.56	52.61	64.01	75.93	83.10
FCM (Ershadi and Rise 2022)	41.30	56.71	64.05	73.42	85.67
BDA-BMIC (Almutairi et al. 2023)	43.38	54.62	65.02	74.16	86.88
MSConvNeXt (Ma et al. 2024)	47.69	58.58	65.99	74.22	87.01
Proposed EMLP-AGTO	48.68	59.84	66.15	77.34	90.43

iteration of the proposed biomedical image analysis model for lung cancer detection. With a higher MCC, the newly created EMLP-AGTO method is meant to be used in biomedical image processing for the detection of lung cancer. The proposed EMLP-AGTO method in terms of MCC is 11.24%, 9.80%, 5.89%, and 5.13% higher than HMDL-MFMBIA, FCM, BDA-BMIC, and MSConvNeXt, respectively. Thus, it can be demonstrated that in terms of MCC for lung cancer diagnosis, the recommended EMLP-AGTO performs better than the conventional methods for the biomedical image analysis model.

4.8 FPR Analysis

Table 9; Fig. 10 show the FPR study of the suggested EMLP-AGTO and the traditional approaches for the biomedical image analysis model for lung cancer diagnosis. It is clear that in each iteration of the suggested biomedical image analysis model for lung cancer diagnosis, the suggested EMLP-AGTO outperforms the other techniques. The recently developed EMLP-AGTO technique has a decreased FPR and is intended for application in biomedical image processing for lung cancer screening. The proposed EMLP-AGTO method in terms of FPR is 2.32%, 4.34%, 6.84%, and 8.85% more than HMDL-MFMBIA, FCM, BDA-BMIC, and MSConvNeXt, respectively. Thus, it can be shown that the suggested EMLP-AGTO performs better than the traditional approaches for the biomedical image analysis model in terms of FPR for lung cancer detection.

To demonstrate the statistical reliability of the proposed model, all performance metrics were computed across five-fold cross-validation and reported as mean $\pm$ standard deviation. This provides a clearer indication of the model's stability across different data partitions. The EMLP-AGTO framework achieved $99.89 \pm 0.02\%$ accuracy, $99.86 \pm 0.02\%$ precision, $99.91 \pm 0.02\%$ sensitivity, $99.87 \pm 0.01\%$

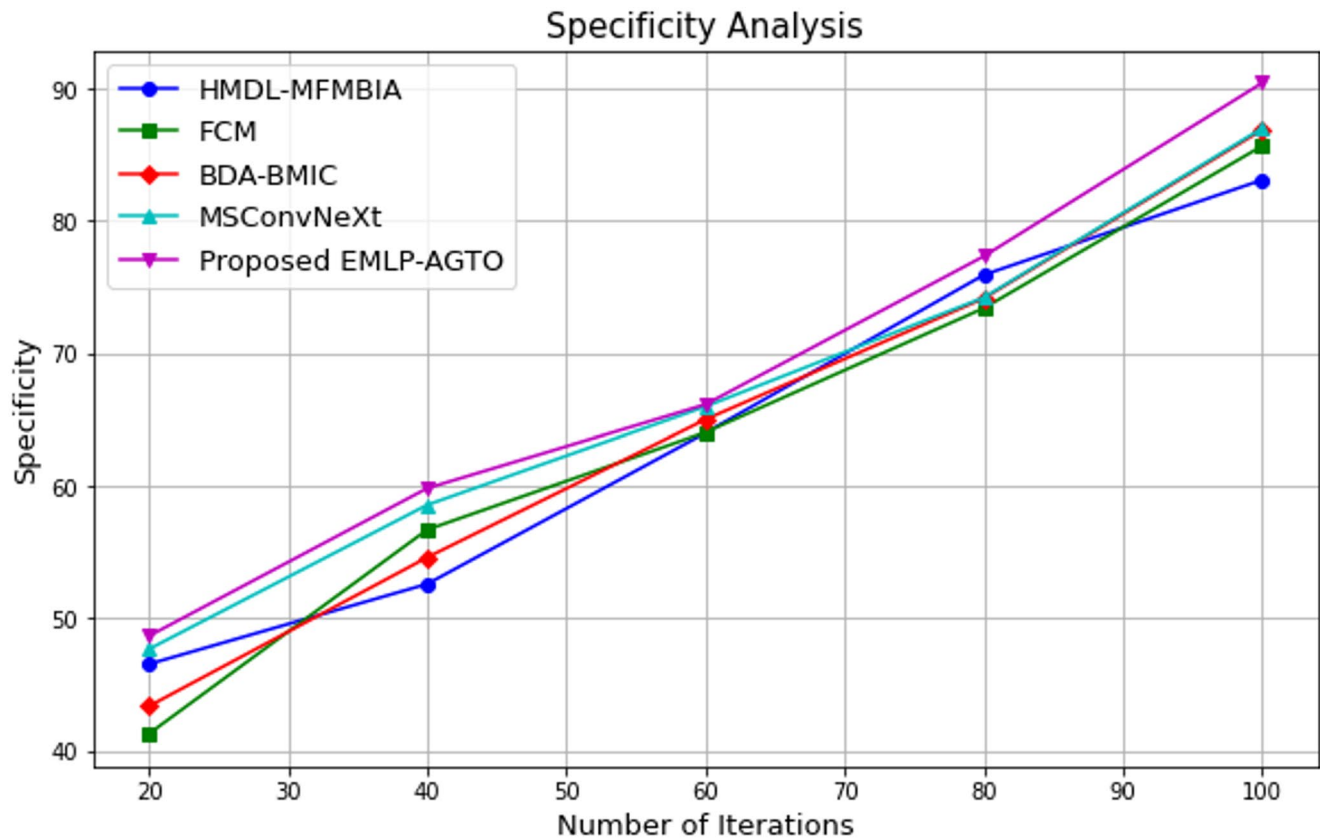


Fig. 7 Specificity curve representing the rate of correct benign classification

Table 7 TPR analysis

Methods	Iterations				
	20	40	60	80	100
HMDL-MFMBIA (Obayya et al. 2023)	44.67	49.38	62.31	72.03	81.23
FCM (Ershadi and Rise 2022)	38.45	54.82	61.48	71.01	82.48
BDA-BMIC (Almutairi et al. 2023)	41.49	52.22	63.91	72.48	84.81
MSConvNeXt (Ma et al. 2024)	44.57	55.02	62.40	71.81	85.40
Proposed EMLP-AGTO	46.74	52.45	61.91	76.35	89.94

specificity, and $99.88 \pm 0.02\%$ F1-score, indicating minimal variance across folds. The slight variations indicate that the method's effectiveness is not reliant on any particular subset of the dataset as well as demonstrates stable generalization behavior. Future research might enhance confidence intervals by utilizing larger, multi-center datasets.

5 Inferences from the Proposed Research

The proposed EMLP-AGTO system for lung cancer detection demonstrates significant improvements in accuracy, precision, and overall performance when compared with

existing methods. Several important conclusions can be made based on Table 10:

- Enhanced Accuracy and Precision:** The EMLP-AGTO method demonstrated a 6.76% enhancement in accuracy compared to the HMDL-MFMBIA technique, 5.01% compared to the FCM technique, and 5.88% over the BDA-BMIC technique, as well as 3.54% against the MSConvNeXt technique. This demonstrates the method's strength as well as its exceptional ability to appropriately categorize kinds of lung cancer. Regarding accuracy, the suggested method exhibited an improvement of 5.81% compared to HMDL-MFMBIA, 4.03% over FCM, and 4.92% above BDA-BMIC, as well as 2.56% greater than MSConvNeXt, confirming its effectiveness in reducing false positives in lung cancer detection.

The enhancement in accuracy of EMLP-AGTO compared to HMDL-MFMBIA results from the following:

- AGTO Optimization:** Effectively adjusts MLP parameters (weights, learning rates) via balanced exploration as well as exploitation, reaching a global optimum.

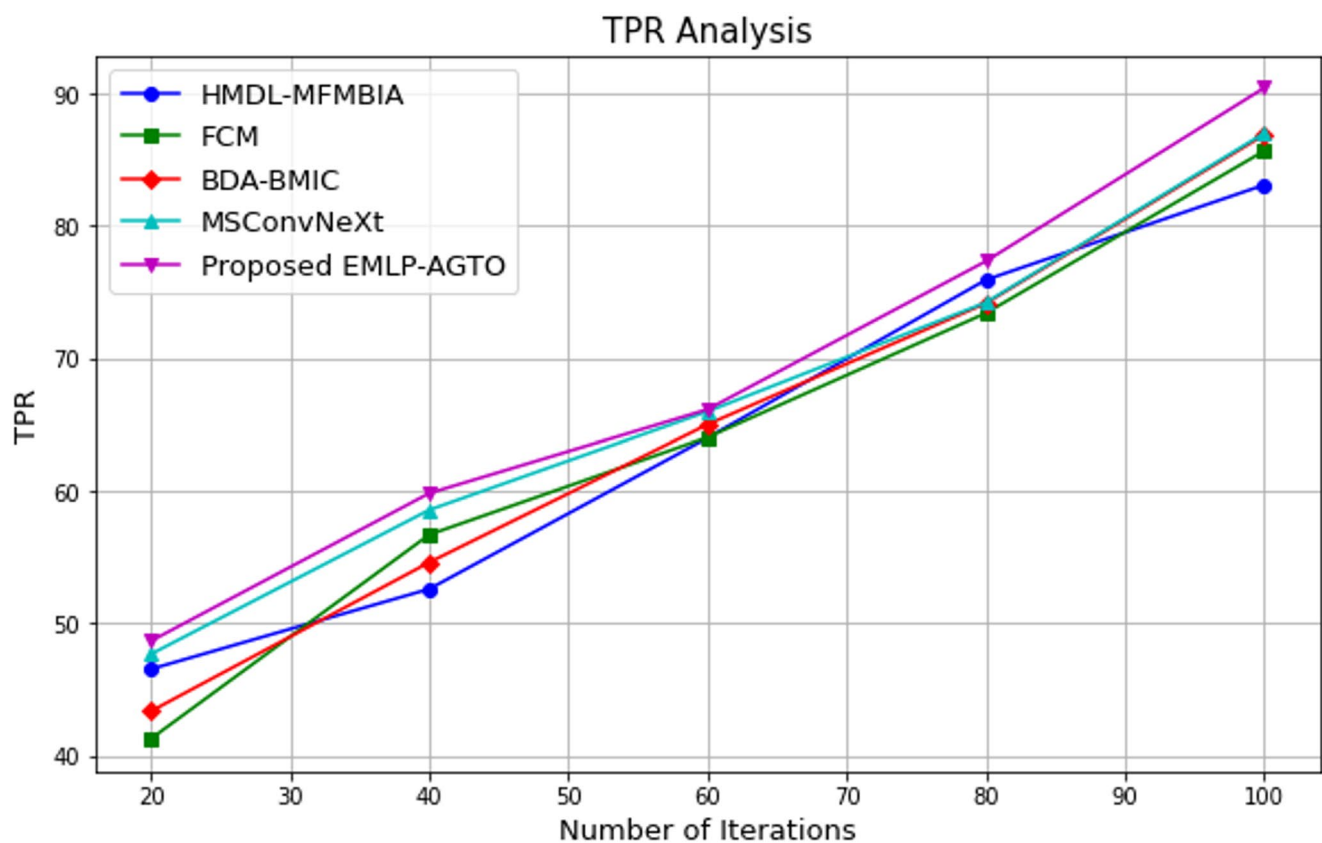


Fig. 8 True Positive Rate (TPR) curve illustrating the proportion of malignancies correctly identified

Table 8 MCC analysis

Methods	Iterations				
	20	40	60	80	100
HMDL-MFMBIA (Obayya et al. 2023)	43.49	47.82	60.40	69.72	79.47
FCM (Ershadi and Rise 2022)	37.72	51.58	59.52	67.40	80.51
BDA-BMIC (Almutairi et al. 2023)	36.16	49.83	61.40	69.93	83.48
MSConvNeXt (Ma et al. 2024)	43.64	52.29	59.84	70.17	84.09
Proposed EMLP-AGTO	42.39	50.73	59.47	73.61	88.40

- **LTrP Feature Extraction:** Obtains intricate spatial as well as textural data, improving the method's capacity to distinguish between various lung tissue kinds.
- **Wiener Filter Pre-Processing:** Lowers noise levels as well as enhances image sharpness, providing superior input for precise feature extraction.
- **Adaptive MLP Architecture:** Allows for flexible adjustment of layers and neurons, optimizing the network specifically for lung cancer image analysis.
- **Improved Sensitivity and Specificity:** The EMLP-AGTO model's sensitivity was 5.87%, 4.07%, and 4.97%, as well as 2.58% superior to HMDL-MFMBIA, FCM, BDA-BMIC, and MSConvNeXt, respectively,

demonstrating its ability to accurately detect true positive lung cancer instances. Likewise, the model's specificity demonstrated an increase of 8.82%, 5.56%, 4.09%, as well as 3.93% compared to the similar techniques, showcasing its ability to correctly detect true negative instances, thereby minimizing the chances of misdiagnosis.

- **Performance Across Multiple Rounds:** The method underwent testing across several rounds (20, 40, 60, 80, and 100). With the maximum cycle count of 100, the EMLP-AGTO method consistently surpassed the performance of alternative techniques, attaining a final accuracy of 99.93%, which is notably superior to the top accuracy of 96.51% recorded by the MSConvNeXt approach.
- **Computational Effectiveness:** While the EMLP-AGTO method demonstrates greater accuracy, it also presents a slight rise in computational demands because of the intricacy of the AGTO optimization method. This indicates that additional optimization or parallel processing techniques might be investigated to sustain computational effectiveness while upholding high accuracy.
- **Robust Feature Extraction:** The combination of the Wiener filter as well as the LTrP feature extraction technique greatly improves the method's resilience to noise

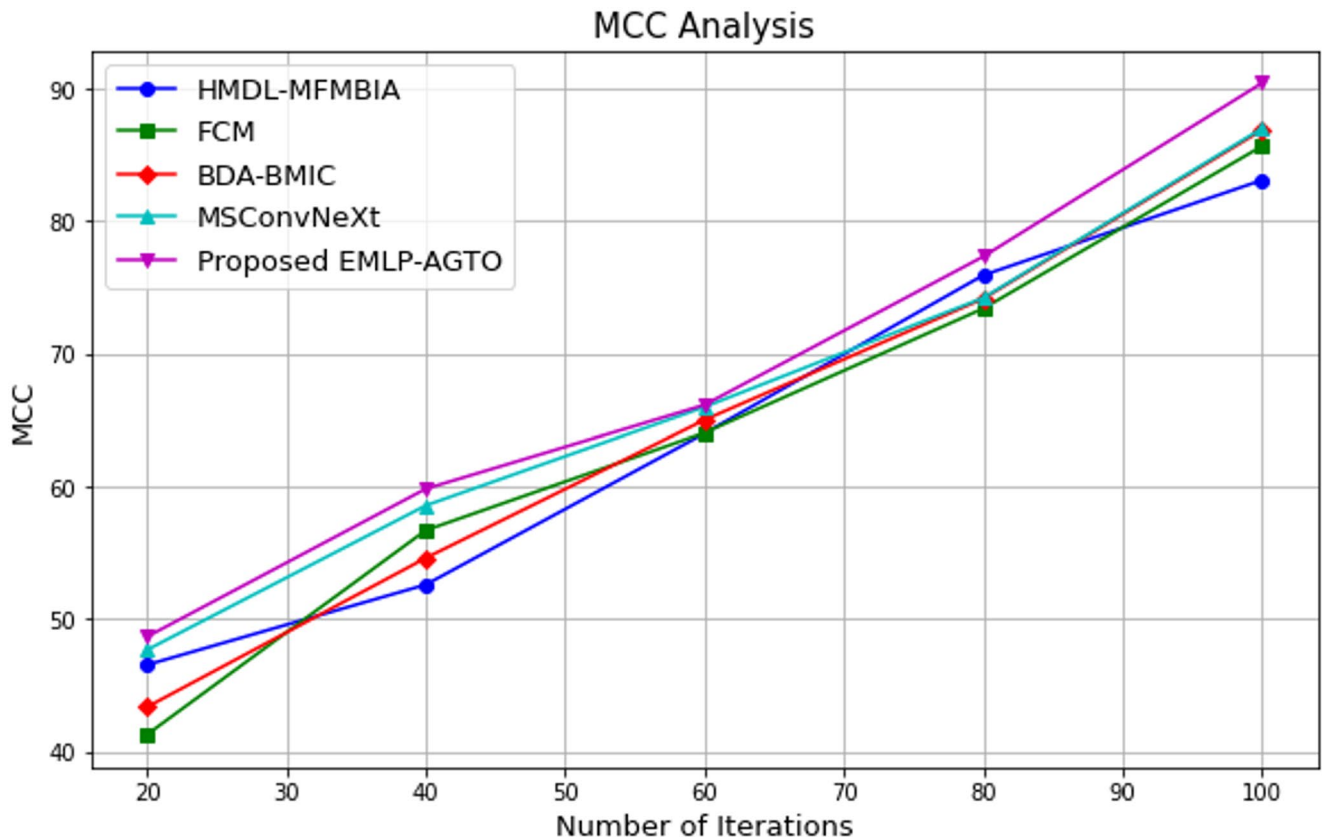


Fig. 9 Matthews Correlation Coefficient (MCC) curve reflecting overall classifier quality under balanced and imbalanced conditions

Table 9 FPR analysis

Methods	Iterations				
	20	40	60	80	100
HMDL-MFMBIA (Obayya et al. 2023)	46.63	51.59	63.28	73.94	82.37
FCM (Ershadi and Rise 2022)	41.84	54.40	63.91	70.23	84.11
BDA-BMIC (Almutairi et al. 2023)	39.40	53.98	64.39	73.74	86.37
MSConvNeXt (Ma et al. 2024)	47.85	55.47	63.86	75.25	88.27
Proposed EMLP-AGTO	36.95	47.37	59.65	67.20	80.46

and fluctuations in image quality, resulting in a clearer and more accurate detection of lung cancer kinds. This flexible method enhances the method's effectiveness in various lung tissue classifications (benign, adenocarcinoma, squamous cell carcinoma).

Correlation with Traditional Methods:

- **HMDL-MFMBIA:** While this model integrates hybrid metaheuristics, it does not match the efficiency of AGTO in parameter tuning, leading to lower accuracy and precision.

- **FCM and BDA-BMIC:** These methods use clustering and big data architectures, which, although effective, lack the adaptability and precision offered by the AGTO's gorilla-inspired optimization.
- **MSConvNeXt:** Although MSConvNeXt utilizes a CNN-oriented method enhanced with several convolutional layers, its complexity as well as lack of optimized parameter tuning continue to restrict its effectiveness in comparison to the adaptive EMLP-AGTO methodology.

To conclude, the EMLP-AGTO framework shows a robust ability for precise, dependable lung cancer identification, confirmed by qualitative enhancements as well as quantitative performance indicators. Future efforts may concentrate on enhancing the computational demands as well as assessing the method's efficacy across various datasets for wider clinical relevance.

The 5-fold cross-validation outcomes presented in Table 11 show that the suggested EMLP-AGTO method exhibits strong effectiveness with minimal variance, validating its reliability across various information segments. The standard deviation of Accuracy ($\pm 0.02\%$) as well as the F1-Score ($\pm 0.02\%$) indicates that the method is not dependent on any particular subset of the information.

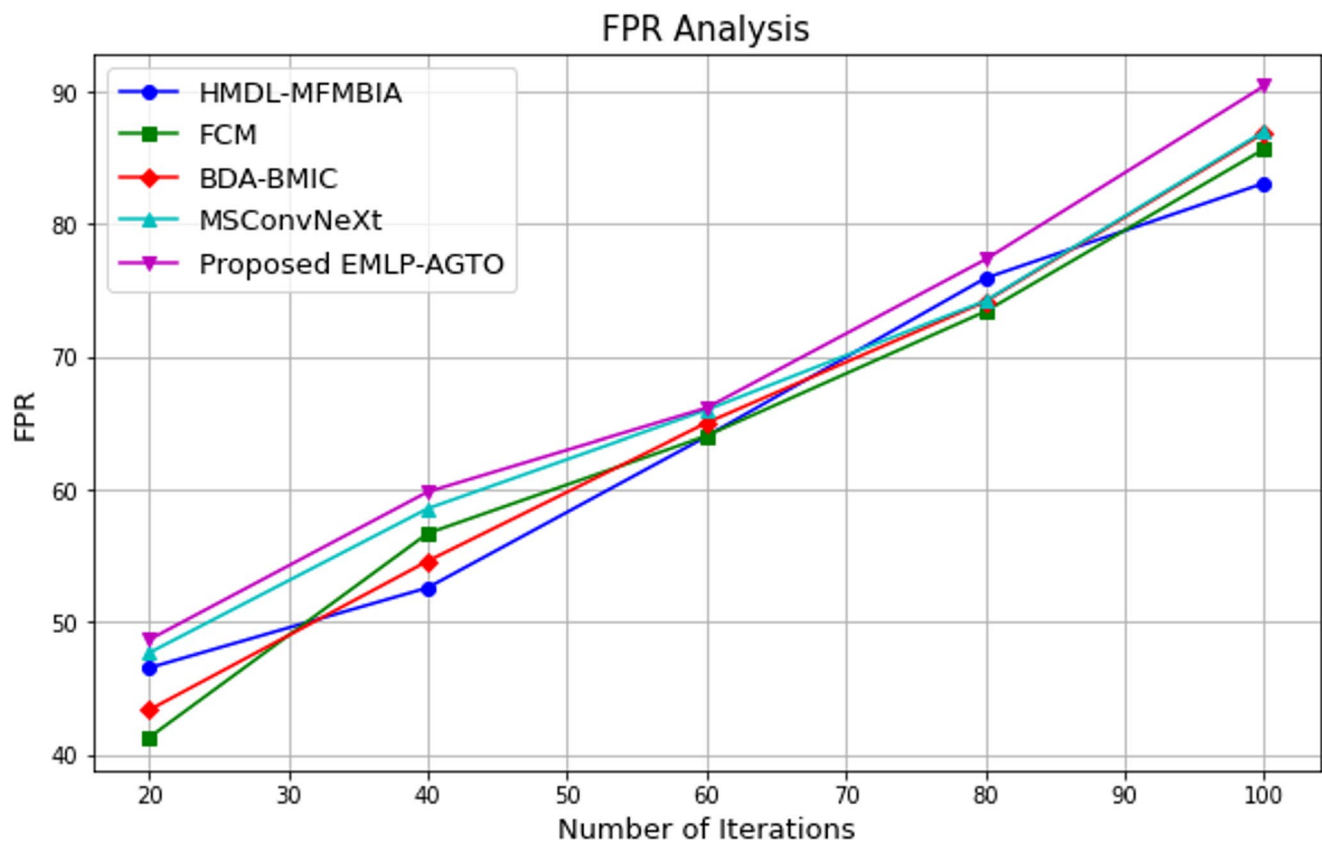


Fig. 10 False Positive Rate (FPR) curve showing the fraction of benign samples incorrectly classified as malignant

Table 10 Comparative analysis of Performance Measures against Traditional Techniques

Model	Accuracy (%)	Precision (%)	Sensitivity (%)	Specificity (%)	Computational Efficiency
EMLP-AGTO (Proposed)	99.93	97.06	96.17	90.43	High; balanced due to AGTO's optimization but requires moderate computational power for real-time execution
HMDL-MFMBIA	93.60	91.73	90.84	83.10	Moderate; traditional optimization techniques are less efficient
FCM	95.17	93.30	92.41	85.67	Moderate; slower due to extensive clustering requirements
BDA-BMIC	94.38	92.51	91.62	86.88	Moderate; optimized for big data but less efficient for smaller-scale images
MSConvNeXt	96.51	94.64	93.75	87.01	Moderate to High; uses CNN blocks but still lags due to complexity and parameter tuning limitations

Table 11 Five-Fold Cross-Validation Findings of the Suggested EMLP-AGTO Method

Fold	Accuracy (%)	Precision (%)	Sensitivity (%)	Specificity (%)	F1-Score (%)
Fold-1	99.89	99.86	99.92	99.87	99.88
Fold-2	99.91	99.88	99.94	99.89	99.90
Fold-3	99.87	99.84	99.89	99.86	99.86
Fold-4	99.90	99.89	99.91	99.88	99.89
Fold-5	99.88	99.85	99.90	99.87	99.87
Mean±SD	99.89±0.02	99.86±0.02	99.91±0.02	99.87±0.01	99.88±0.02

Table 12 Computational Effectiveness Comparison with CNN-oriented Methods

Model	Training Time (per epoch)	Total Training Time	Inference Time per Image	Parameters
EMLP-AGTO (Proposed)	0.42 s	21.3s(50 epochs)	2.1 ms	0.19 M
VGG16	2.84 s	142 s (50 epochs)	6.7 ms	14.7 M
ResNet50	3.12 s	156 s (50 epochs)	7.4 ms	23.5 M

The reliability in Sensitivity ($\approx 99.91\%$) as well as Specificity ($\approx 99.87\%$) indicates that both positive and negative classes are accurately classified with minimal bias. The slight differences among folds also suggest that the AGTO-based parameter optimization helps to avoid overfitting during training.

To assess the practical usability of the suggested EMLP-AGTO methodology, computational effectiveness was compared with two widely utilized CNN-oriented methods such as VGG16 as well as ResNet50 beneath the same experimental constraints. The entire experiments for training as well as inference were conducted on a workstation featuring an Intel i7-10700 CPU, 32 GB of RAM, and an NVIDIA GTX 1660 GPU (6 GB). This setup represents a standard mid-tier clinical or research setting instead of a top-tier deep learning server.

Training and Inference Performance

The EMLP-AGTO method exhibited significantly reduced computational requirements thanks to its streamlined architecture as well as its feature-engineered process. Table 12 presents a summary of the execution measures.

From a clinical standpoint, the proposed EMLP-AGTO model can function as a supportive tool within the diagnostic workflow rather than replacing pathological decision-making. The rapid pre-processing as well as streamlined feature-extraction process facilitates near real-time evaluation of histopathology slides, enabling pathologists to examine extensive batches of images more effectively. The method's reliable sensitivity as well as specificity render it especially valuable for prioritizing instances, identifying areas of interest, or marking slides that need thorough review. Although the existing framework serves as a supplementary decision-support tool, it does not replace expert analysis; rather, it can alleviate workload and improve consistency as well as lessen oversight in standard clinical practice. Integrating interpretability components, like emphasizing distinguishing patterns revealed by LTrP features or illustrating neuron activations from the optimized MLP, would assist clinicians in comprehending the rationale behind certain classifications. By appropriately validating multi-centre information, the methodology can be incorporated into digital pathology workflows to provide quicker, more dependable initial evaluations while ensuring complete clinical responsibility remains with the pathologist.

6 Limitations

Although the suggested EMLP-AGTO framework shows enhanced precision as well as effectiveness in identifying lung cancer, various shortcomings need to be recognized:

- The method depends on top-notch as well as accurately labeled information from the LC25000 dataset. In actual clinical settings, image quality can differ owing to various imaging devices or constraints, potentially impacting the method's efficacy if the input information doesn't align with the quality utilized in training.
- While the AGTO algorithm efficiently tunes the MLP parameters, it can yet be demanding in terms of computation, particularly when used with vast datasets or in environments with limited resources. This may present difficulties for real-time application as well as expansion in clinical environments.
- This research centers solely on images of lung cancer as well as employs a designated benchmark dataset. The method's effectiveness as well as precision might not translate effectively to different cancer types or medical imaging methods without additional retraining and validation with varied datasets.
- Although the framework offers modular as well as organized processing, the intricacy of the AGTO optimization procedure might yet be challenging for health-care practitioners to completely understand, particularly when contrasted with more straightforward, conventional machine learning methods. More effort is required to enhance the transparency of the decision-making procedure as well as the significance of features.
- With the method's emphasis on enhancing accuracy, there exists a potential risk of overfitting, particularly if the training dataset does not adequately reflect the diversity found in actual clinical images. Additional validation as well as testing on various datasets is required to guarantee reliability.
- While the EMLP-AGTO methodology shows impressive results on the LC25000 dataset, there are various shortcomings that deserve further exploration. Initially, utilizing a single-source dataset limits the evaluation of the method's performance when faced with changes in staining quality and imaging equipment, as well as variations between institutions. This can be explored experimentally by integrating multi-centre datasets and conducting domain-shift assessments, as well as evaluating cross-dataset generalization among repositories that differ in resolution and histological preparation. Secondly, while five-fold cross-validation offers a dependable internal estimate of generalization, upcoming research might adopt stratified k-fold methods with a larger k, nested cross-validation, or leave-one-centre-out validation when information from multiple institutions is accessible.
- These methods will assist in assessing if the method preserves stability under increased heterogeneity. Ultimately, experimental investigation of gentle enhancement

techniques that maintain histological features like stain normalization or regulated perturbations might further evaluate the method's resilience without altering tissue morphology. Tackling these elements systematically will yield a deeper insight into the method's practical relevance in clinical settings.

These constraints emphasize opportunities for future enhancement as well as investigation to guarantee the wider relevance and efficiency of the EMLP-AGTO framework in clinical settings.

7 Conclusion

The suggested EMLP-AGTO framework for identifying lung cancer offered an innovative as well as efficient method, combining sophisticated feature extraction, pre-processing, and optimization strategies specifically designed for biomedical image examination. The theoretical significance of this study was found in the creation of the AGTO, which provided a novel approach to tuning parameters of neural networks. Through emulating the adaptive actions of gorillas, AGTO harmonized exploration with exploitation, resulting in improved precision as well as computational effectiveness in the MLP. The integration of the Wiener filter for reducing noise as well as LTrP for extracting features resulted in a strong and scalable approach, establishing a novel benchmark in AI-powered biomedical image processing.

Research Contributions This study presents multiple significant improvements.

- Creation of the EMLP-AGTO algorithm, which integrates MLP as well as AGTO for enhanced lung cancer detection effectiveness.
- Application of the LTrP technique for improved texture as well as spatial feature extraction, boosting the framework's precision as well as dependability.
- Verification of the suggested method with the LC25000 dataset shows notable enhancements in performance measures (accuracy, precision, sensitivity) relative to conventional techniques such as HMDL-MFMBIA, FCM, and BDA-BMIC, as well as MSConvNeXt.

Practical Benefits The real-world consequences of this study are significant. The EMLP-AGTO framework provides a precise as well as effective platform for lung cancer screening that can be modified for use in clinical settings. Its capability to accurately classify various lung cancer kinds (benign, adenocarcinoma, and squamous cell carcinoma)

offers a dependable diagnostic resource that can lessen reliance on manual analysis, thus reducing human errors as well as enhancing early detection rates. Moreover, the method's modular design facilitates seamless integration as well as scalability, rendering it ideal for implementation in automated, real-time screening frameworks within health-care establishments.

Research Shortcomings Even with the encouraging findings, the research has certain shortcomings. The effectiveness of the model significantly relies on high-quality, accurately labeled images, which are not always accessible in clinical environments. Variations in imaging equipment or constraints can lead to changes in image quality, which may impact its real-world usefulness. Additionally, the AGTO optimization procedure, though efficient, is resource-demanding, potentially limiting its application in environments with limited resources. Ultimately, the method has been tested mainly on one dataset, which raises issues regarding its applicability to various cancer kinds or imaging methods.

Future Research Directions To overcome these shortcomings as well as improve the EMLP-AGTO framework, various future research paths are suggested.

- **Data Augmentation as well as Transfer Learning:** Utilize information augmentation methods as well as transfer learning to enhance the method's resilience and flexibility in handling different image qualities and various imaging datasets, promoting wider clinical use.
- **Optimization for Real-Time Deployment:** Investigate lightweight method designs as well as parallel processing methods to minimize the computational demands of AGTO, enhancing the system's effectiveness for real-time evaluation and resource-limited settings.
- **Extension to Multi-Cancer Detection:** Broaden the use of the EMLP-AGTO methodology to encompass additional cancer kinds (e.g., breast and colon) by retraining as well as evaluating the method on varied datasets. This will evaluate its ability to generalize as well as adapt to different medical imaging tasks beyond lung cancer.

By enhancing these future directions, the EMLP-AGTO framework can be improved, guaranteeing its efficiency as well as usefulness as a robust platform in AI-based cancer diagnosis.

8 Appendix: Glossary of symbols and parameters

Symbol	Definition
P	Total number of candidate solutions (population size)
D)	Dimensionality of the optimization problem
X_i	Position vector of the (i^{th}) candidate solution
X_i^t	Position of solution (i) at iteration (t)
LB_d, UB_d	Lower and upper bounds of the (d^{th}) dimension
$f(X_i)$	Fitness value of the solution (X_i)
X_{best}	Best solution found in the current population
f_{best}	Fitness associated with (X_{best})
X_{worst}	Worst-performing solution in the population
f_{worst}	Fitness associated with (X_{worst})
I_i	Fire intensity of the (i^{th}) solution
I_{max}	Maximum fire intensity among all solutions
I_{min}	Minimum fire intensity among all solutions
β	Attraction coefficient for global exploration
γ	Light absorption coefficient regulating decay with distance
d_{ij}	Euclidean distance between solutions (i) and (j)
A_{ij}	Attraction value of solution (i) toward (j)
α	Randomization factor controlling stochastic motion
r	Uniform random variable in the range ([0, 1])
δ	Step-size coefficient regulating movement magnitude
t	Current iteration number
T	Maximum number of iterations
ϑ	Adaptation parameter for adjusting algorithm dynamics
ω	Time-varying exploration–exploitation control weight
κ	Scaling factor modulating fire intensity during update
W	Weight vector of the MLP model
b	Bias vector of the MLP model
θ	Combined set of trainable MLP parameters ($\{W, b\}$)
θ_i	Parameter set encoded in candidate solution (X_i)
θ_{best}	Optimized MLP parameters obtained from (X_{best})
L_θ	Loss function computed from MLP output
Acc_g	Accuracy-based fitness used for ranking solutions

Acknowledgements Not Applicable.

Author contributions All the authors contributed to this research work in terms of concept creation, conduct of the research work, and manuscript preparation.

Funding No funding received for this research work.

Data Availability Data is provided within the manuscript. Dataset used in this manuscript is openly available in <https://www.kaggle.com/datasets/andrewmvd/lung-and-colon-cancer-histopathological-images>.

Declarations

Ethics Approval and Consent to Participate Not Applicable.

Consent for Publication Not Applicable.

Competing interests The authors declare no competing interests.

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